# VU MEDICAL ZONE

## AMAAN KHAN

## ZOO 501 – DEVELOPMENTAL BIOLOGY

### LECTURE 73 TO 144

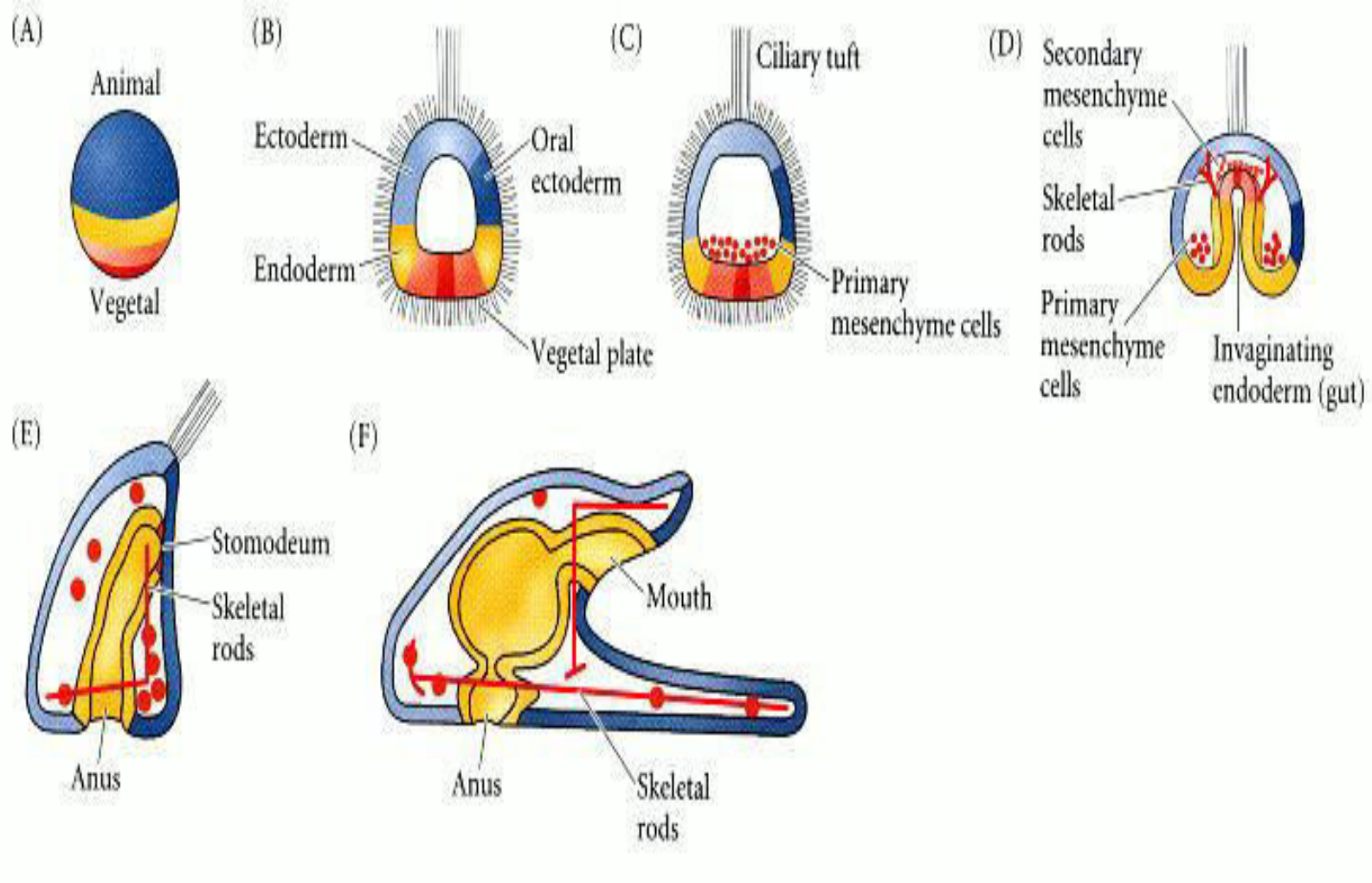
FINAL TERM PPT



# Sea Urchin Gastrulation

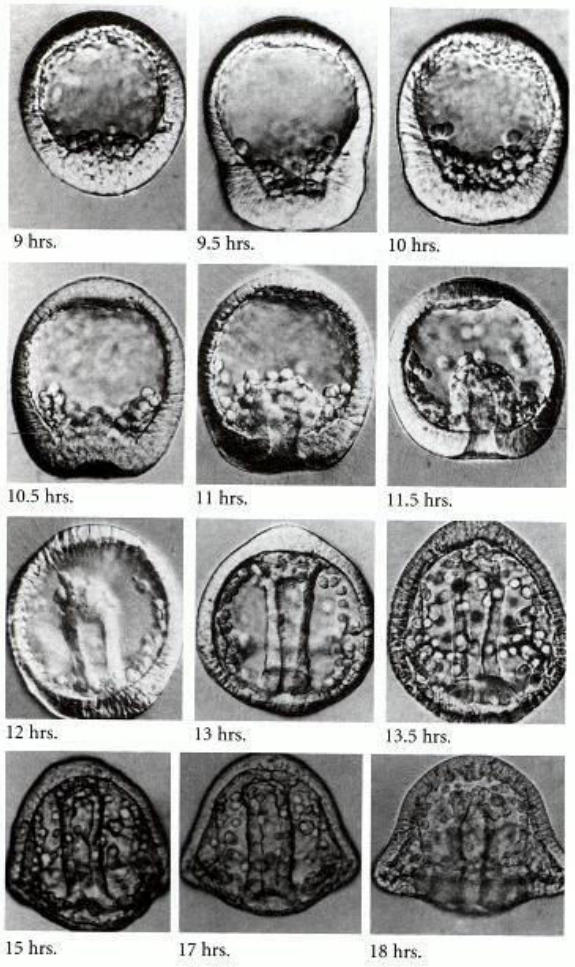
# Sea Urchin Gastrulation

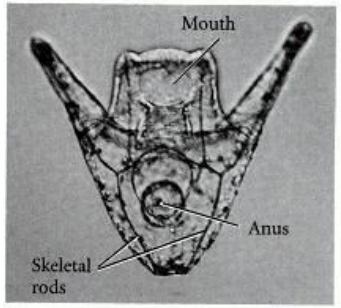
Figure show the fates of the various regions of the blastula as it develops through gastrulation to the Pluteus larva stage characteristic of sea urchins.



# Sea Urchin Gastrulation

Shortly after the blastula hatches from its fertilization envelope, the vegetal side of the spherical blastula begins to thicken and flatten. At the centre of this flat vegetal plate, a cluster of small cells begins to change. These cells begin extending and contracting long, thin (30 × 5 µm) processes called **filopodia** from their inner surfaces.



# Sea Urchin Gastrulation

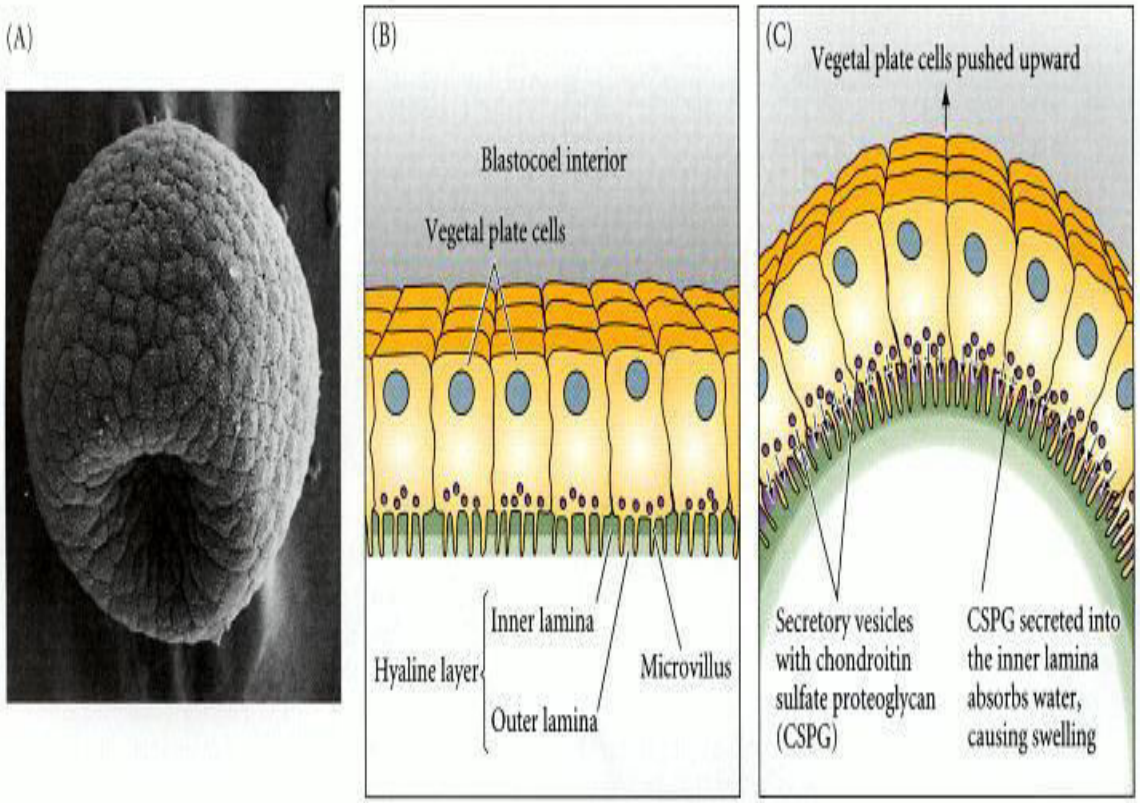
These cells, derived from the micromeres, are called the **primary mesenchyme.** They will form the larval skeleton, so they are sometimes called the **skeletogenic mesenchyme.**

# Sea Urchin Gastrulation II

# Sea Urchin Gastrulation

The vegetal plate bends inward and invaginates about one-fourth to one half the way into the blastocoel. Then invagination suddenly ceases. The invaginated region is called the archenteron (primitive gut), and the

region is called the **blas**

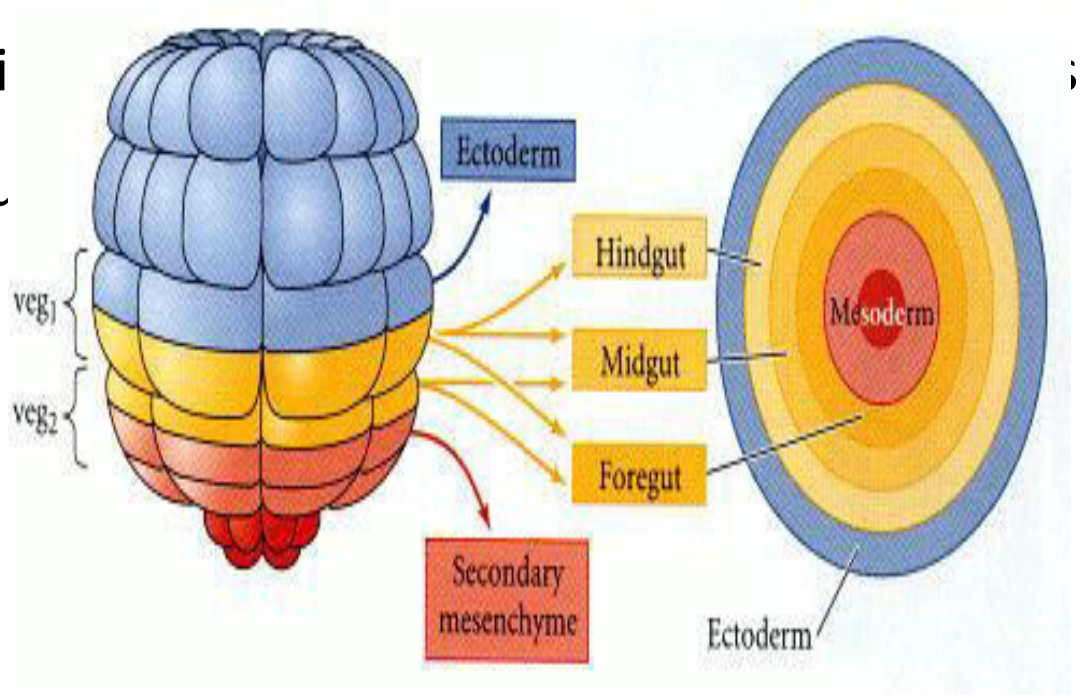


# Sea Urchin Gastrulation

The hyaline layer is actually made up of two layers, an outer lamina made primarily of **hyalin protein** and an inner lamina composed of fibropellin proteins.

# Sea Urchin Gastrulation

The endodermal cells adjacent to the micromere-derived mesenchyme become foregut, migrating the farthest distance into the blastocoel. The next layer of endodermal cells becomes midgut, and the last ci ; the hindgu



# The Early Development of Snails
## Cleavage in Snail Eggs

# Cleavage in Snail Eggs

Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most molluscs. It differs from radial cleavage in numerous ways..

# Cleavage in Snail Eggs

First, the cleavage planes are not parallel or perpendicular to the animal-vegetal axis of the egg; rather, cleavage is at oblique angles, forming a "spiral" arrangement of daughter blastomeres.

# Cleavage in Snail Eggs

Second, the cells touch one another at more places than do those of radially cleaving embryos. In fact, they assume the most thermodynamically stable packing orientation, much like that of adjacent soap bubbles.

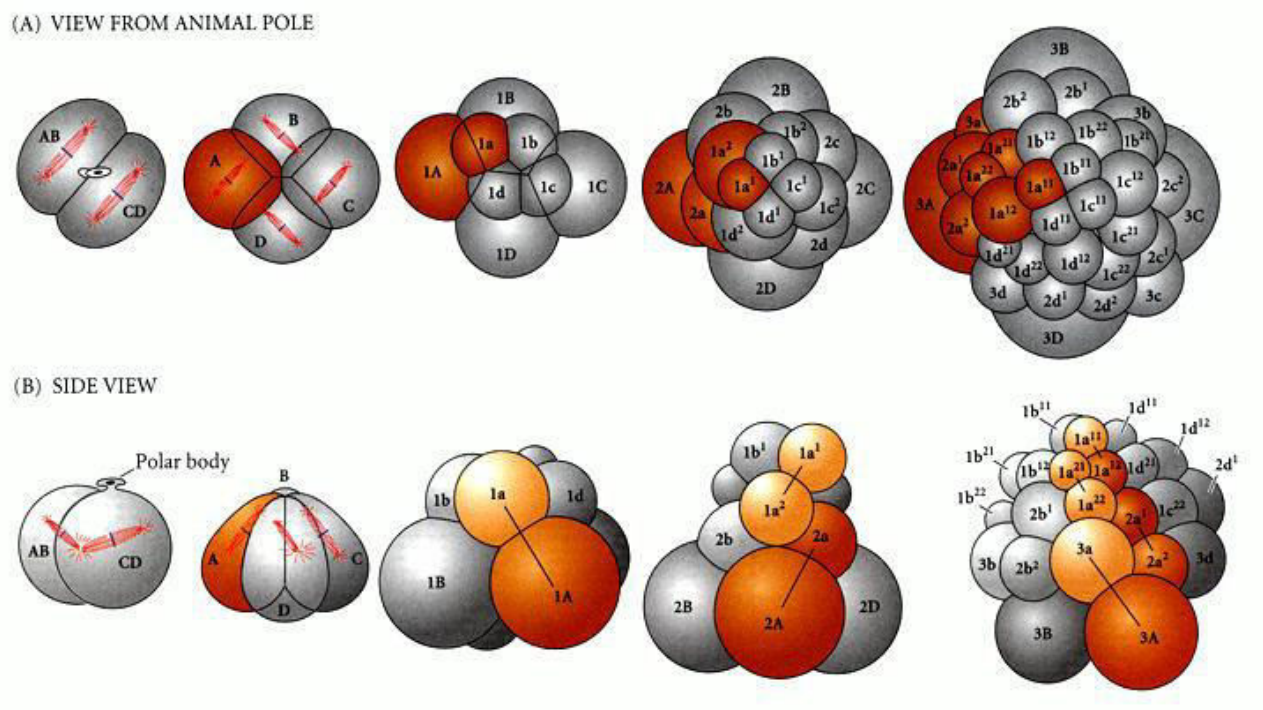
# Cleavage in Snail Eggs

Third, spirally cleaving embryos usually undergo fewer divisions before they begin gastrulation, making it possible to follow the fate of each cell of the blastula.

Blastulae produced by radial cleavage have no blastocoel and are called **stereoblastulae**.

# Cleavage in Snail Eggs

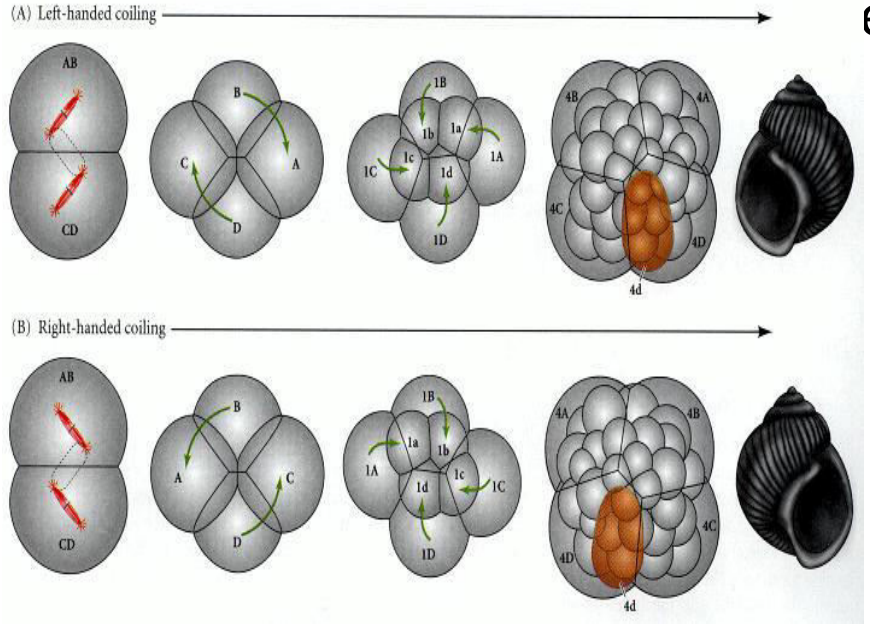
Figure depicts the cleavage of mollusc embryos. The first two cleavages are nearly meridional, producing four large macromeres (labelled A, B, C, and D). In each successive cleavage, each macromere buds off a small micromere at its animal pole.



# Sinistral coiling in Snail

# Cleavage in Snail Eggs

The orientation of the cleavage plane to the left or to the right is controlled by cytoplasmic factors within the oocyte. Some snails have their coils opening to the right of their shells (dextral coiling), whereas other snails ha eft (sinistral coiling).



# Cleavage in Snail Eggs

Experiments have demonstrated that the non-diffusible polar lobe cytoplasm is extremely important in normal mollusc development for a number of reasons.

# Cleavage in Snail Eggs

- It contains the determinants for the proper cleavage rhythm and cleavage orientation of the D blastomere.
- It contains certain determinants (those entering the 4d blastomere and hence leading to the **mesentoblasts**) for mesodermal and intestinal differentiation.

# Cleavage in Snail Eggs

- It is responsible for permitting the inductive interactions (through the material entering the 3d blastomere) leading to the formation of the shell gland and eye.
- It contains determinants needed for specifying the dorsal-ventral axis of the embryo.

# Gastrulation in Snails

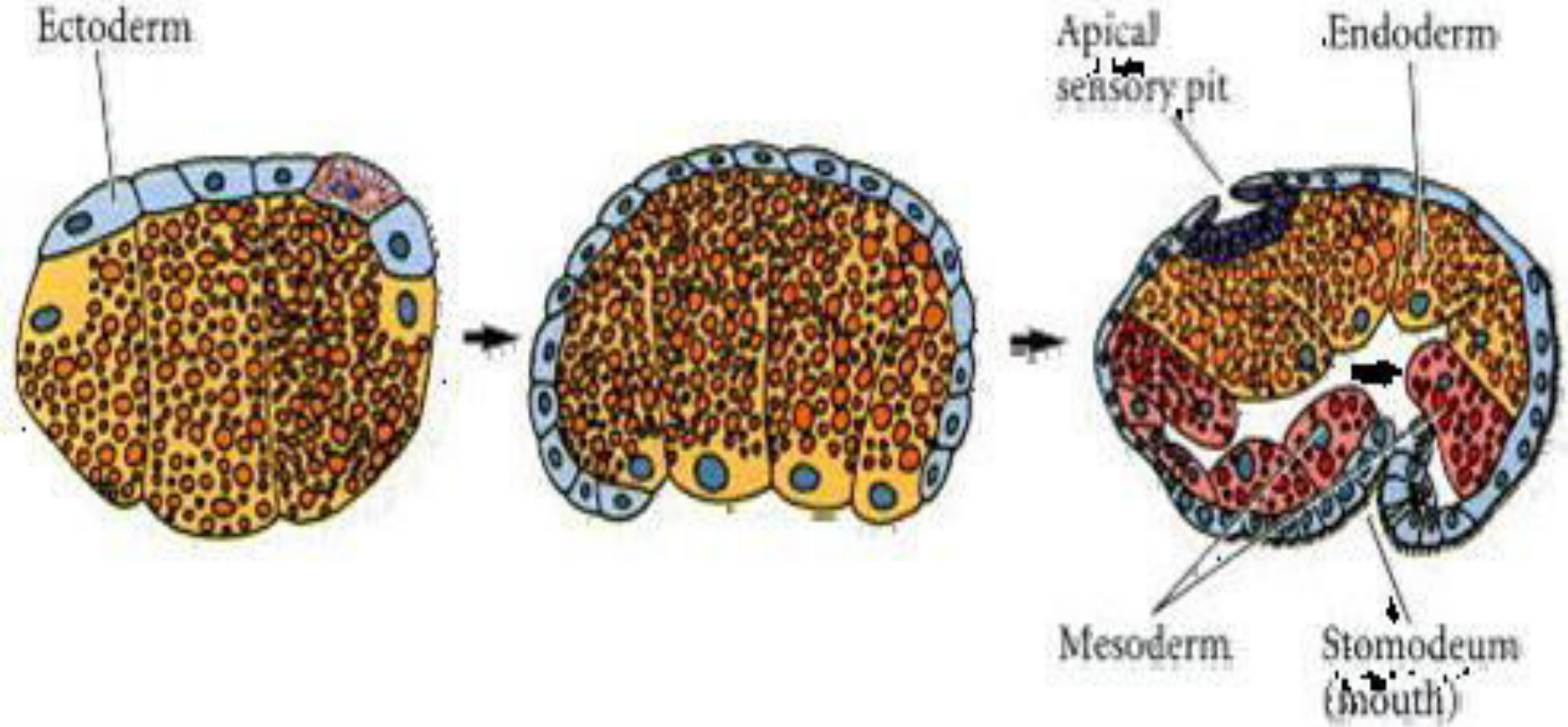
# Gastrulation in Snails

The snail stereoblastula is relatively small.

Gastrulation in Snails is accomplished primarily by epiboly, wherein the micromeres at the animal cap multiply and "overgrow" the vegetal macromeres.

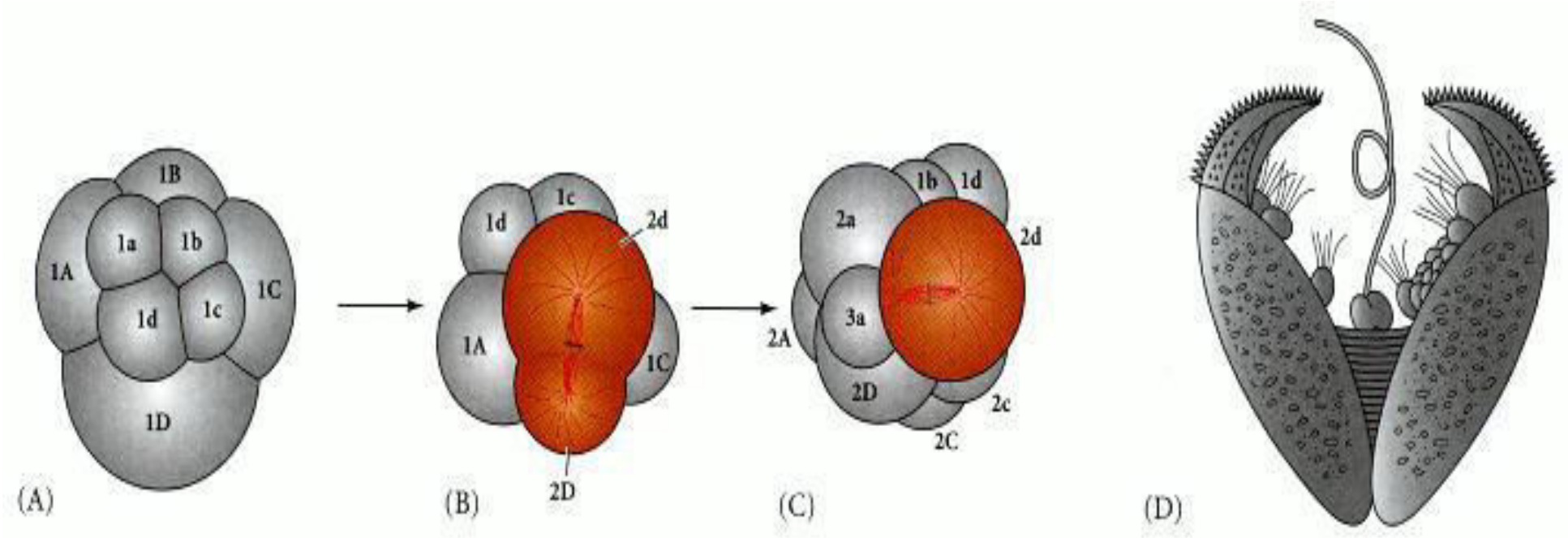
# Gastrulation in Snails

Eventually, the micromeres will cover the entire embryo, leaving a small slit at the vegetal pole.



# Gastrulation in Snails

In the typical cleavage of molluscs, cells divides to produce most of the larval structures, including a gland capable of producing a large shell.



# Shell formation during gastrulation in snails

# Gastrulation in Snails

The **Shell gland** is an ectodermal organ formed through induction by mesodermal cells. Without the mesoderm, no cells are present to induce the competent ectoderm. Here we see an example of limited induction within a mosaic embryo.

# Gastrulation in Snails

The resulting larvae (called **glochidia**) resemble tiny bear traps; they have sensitive hairs that cause the valves of the shell to snap shut when they are touched by the gills or fins of a wandering fish.

# Gastrulation in Snails

They attach themselves to the fish and "hitchhike" with it until they are ready to drop off and metamorphose into adult clams. In this manner, they can spread upstream.

# Early Development in Tunicates
## Tunicate Cleavage

# Tunicate Cleavage

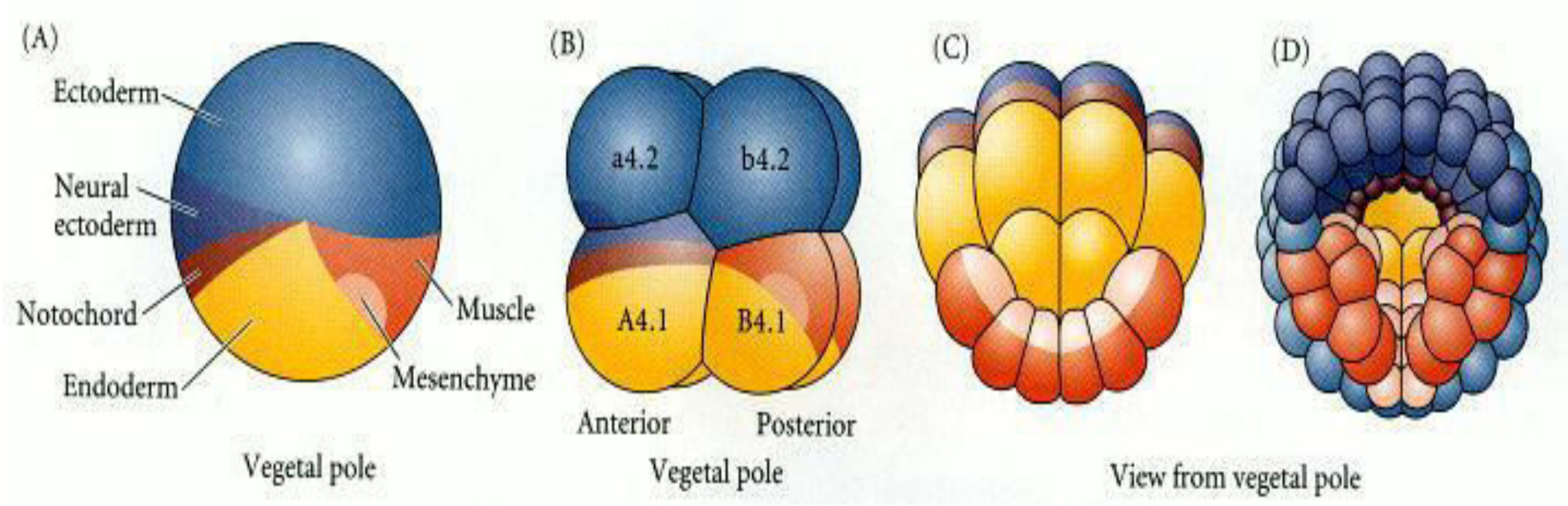
Ascidians, members of the tunicate subphylum, are fascinating animals for several reasons, but the foremost is that they are invertebrate chordates. They have a notochord as larvae (and therefore are chordates), but they lack vertebrae.

# Tunicate Cleavage

As larvae, they are free-swimming tadpoles; but when the tadpole undergoes metamorphosis, it sticks to the sea floor, its nerve cord and notochord degenerate, and it secrete a cellulose tunic (which gave the name "tunicates" to these creatures).

# Tunicate Cleavage

These animals are characterized by bilateral holoblastic cleavage, a pattern found primarily in tunicates.



# Blastula formation in Tunicate

# Tunicate Cleavage

Each successive division orients itself to this plane of symmetry, and the half-embryo formed on one side of the first cleavage plane is the mirror image of the half-embryo on the other side. The second cleavage is meridional, like the first, but unlike the first division, it does not pass through the centre of the egg.

# Tunicate Cleavage

Rather, it creates two large anterior cells (the A and D blastomeres) and two smaller posterior cells (blastomeres B and C). Each side now has a large and a small blastomere.

# Tunicate Cleavage

During the next three divisions, differences in cell size and shape highlight the bilateral symmetry of these embryos. At the 32-cell stage, a small blastocoel is formed, and gastrulation begins.

# Gastrulation in Tunicates

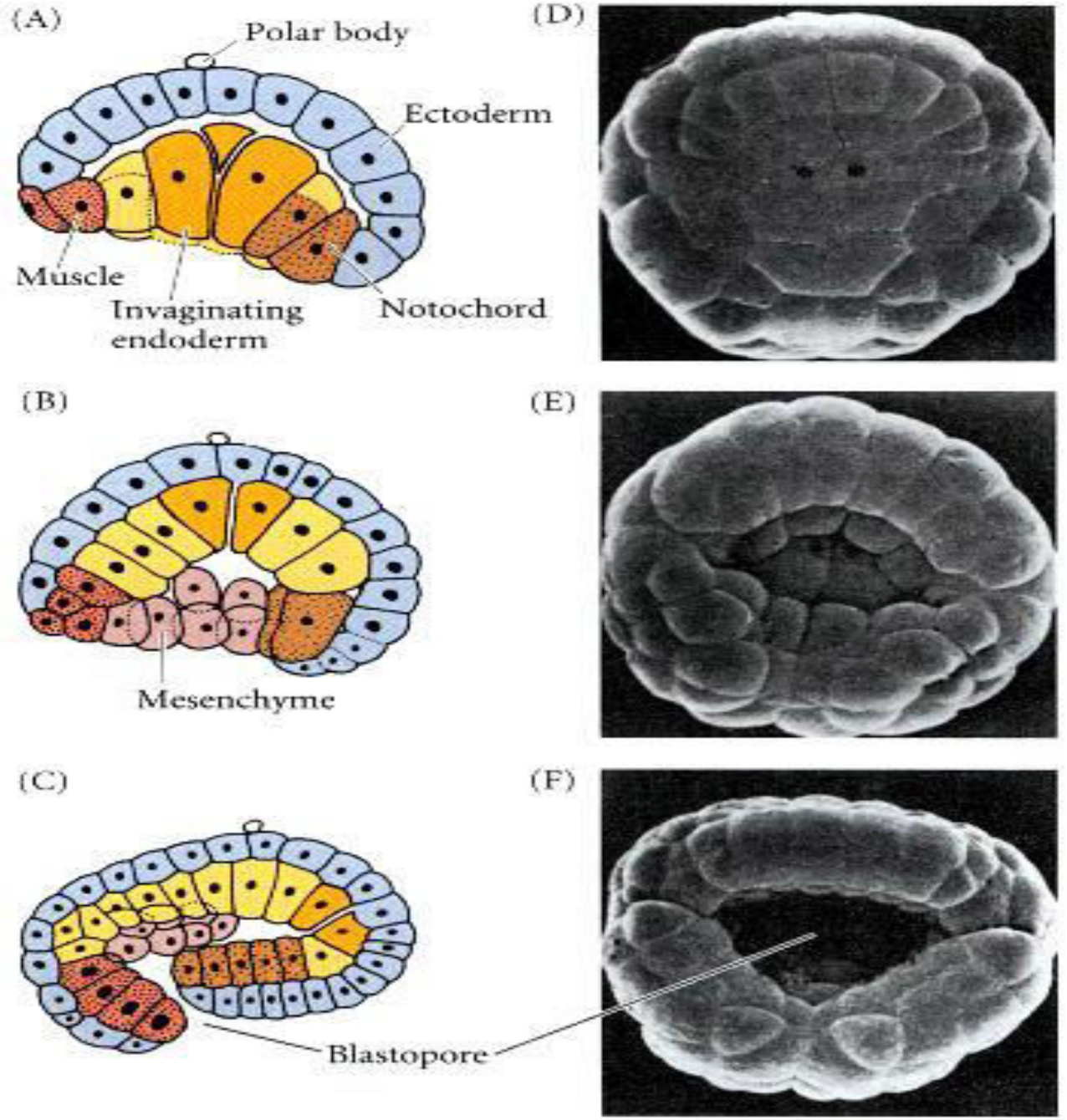
# Gastrulation in Tunicates

Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the mesoderm, and the epiboly of the ectoderm.

About 4 5 hours after fertilization, the vegetal (endoderm) cells assume a wedge shape, expanding their apical margins and contracting near their vegetal margins.

# Gastrulation in Tunicates

Pairs of blastomeres appear to lead this invagination into the centre of the embryo. This invagination forms a blastopore whose lips will become the mesodermal cells.



# Gastrulation in Tunicates

The presumptive notochord cells are now on the anterior portion of the blastopore lip, while the presumptive tail muscle cells (from the yellow crescent) are on the posterior lip of the blastopore.

The lateral lips of the blastopore comprise those cells that will become mesenchyme.

# Gastrulation in Tunicates II

# Gastrulation in Tunicates

The second step of gastrulation involves the involution of the mesoderm. The presumptive mesoderm cells involute over the lips of the blastopore, and by migrating upon the basal surfaces of the ectodermal cells, move inside the embryo.

# Gastrulation in Tunicates

The dorsal ectodermal cells that are the precursors of the neural tube invaginate into the embryo and are enclosed by neural folds. This forms the neural tube, which will form a brain anteriorly and a spinal chord posteriorly.

# Gastrulation in Tunicates

Meanwhile, the presumptive notochord cells on the right and left sides of the embryo migrate to the midline and interdigitate to form the notochord, a single row of 40 cells.

The muscle cells of the tail differentiate on either side of the neural tube and notochord.

# Summary

# Early Invertebrate Development

# Summary
# Early Invertebrate Development

- During cleavage, most cells do not grow. Rather, the volume of the oocyte is cleaved into numerous cells. The major exceptions to this rule are mammals.
- The blastomere cell cycle is governed by the synthesis and degradation of cyclin. Cyclin synthesis promotes the formation of MPF, and MPF promotes mitosis.

# Summary
# Early Invertebrate Development

- A blastomere is a cell derived from cleavage in an early embryo. A blastula is an embryonic structure composed of blastomeres. The cavity in the blastula is the blastocoel. A mammalian blastula is called a blastocyst.

# Summary
# Early Invertebrate Development

- The movements of gastrulation include invagination, involution, ingression, delamination, and epiboly.
- Three axes are the foundations of the body: the anterior-posterior axis (head to tail or mouth to anus), the dorsal-ventral axis (back to belly), and the right-left axis (between the two lateral sides of the body).

# Summary of Invertebrate cleavage

# Summary of Invertebrate cleavage

- In all four invertebrates described here, cleavage is holoblastic. In the sea urchin, cleavage is radial; in the snail, spiral; in the tunicate, bilateral; and in the nematode, rotational.
- In the tunicate, snail, and nematode, gastrulation occurs when there are relatively few cells, and the blastopore becomes the mouth. This is the protostome mode of gastrulation.

# Summary of Invertebrate cleavage

- In the sea urchin, gastrulation occurs only after thousands of cells have formed, and the blastopore becomes the anus. This is the deuterostome mode of gastrulation, and is characteristic only of echinoderms and chordates.
- In sea urchins, cell fates are determined by signalling. The micromeres constitute a major signalling centre. β-catenin is important for the inducing capacity of the micromeres.

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# Early Development of Vertebrates
## Cleavage in Fishes

# Cleavage in Fish

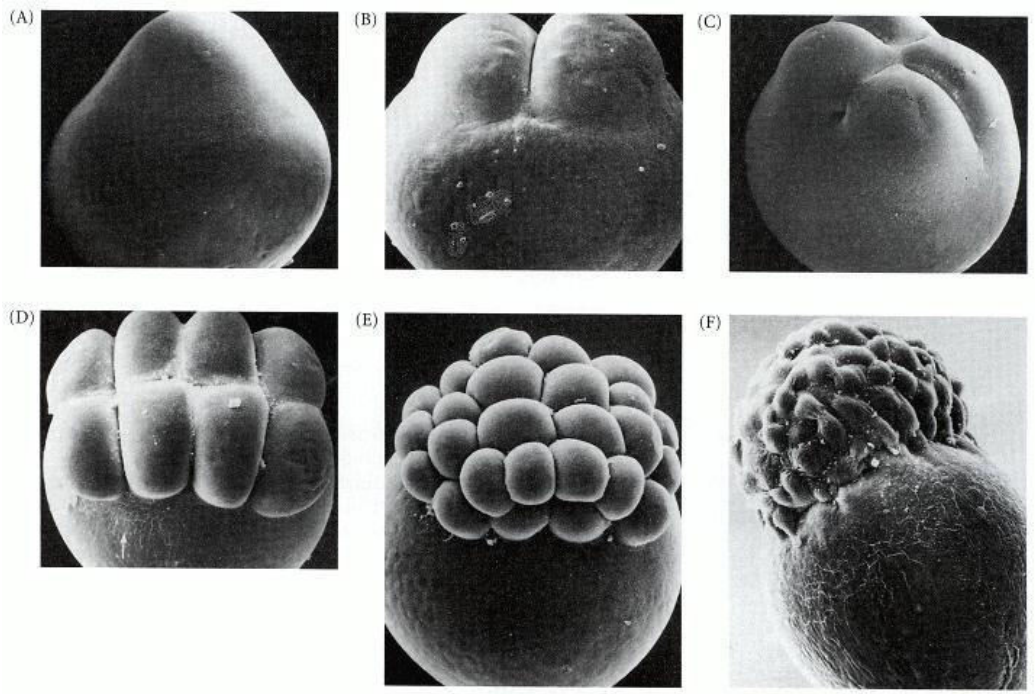
In fish eggs, cleavage occurs only in the **blastodisc**, a thin region of yolk-free cytoplasm at the animal cap of the egg. Most of the egg cell is full of yolk. The cell divisions do not completely divide the egg, so this type of cleavage is called meroblastic.

# Cleavage in Fish

Since only the cytoplasm of the blastodisc becomes the embryo, this type of meroblastic cleavage is called discoidal. The calcium waves initiated at fertilization stimulate the contraction of the actin cytoskeleton to squeeze non-yolky cytoplasm into the animal pole of the egg.

# Cleavage in Fish

Scanning electron micrographs show beautifully the incomplete nature of discoidal meroblastic cleavage in fish eggs.



# Cleavage in Fish

These divisions are rapid, taking about 15 minutes each. The first 12 divisions occur synchronously, forming a mound of cells that sits at the animal pole of a large yolk cell. These cells constitute the blastoderm.

# Cleavage in Fish II

# Cleavage in Fish

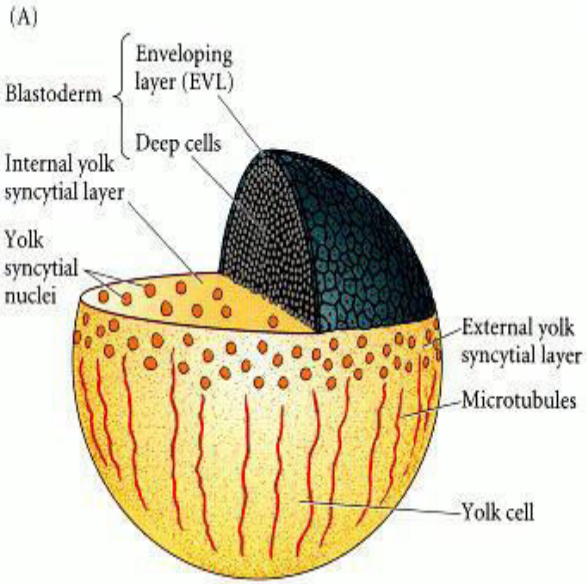
Three distinct cell populations can be distinguished. The first of these is the **yolk syncytial layer (YSL)**. The YSL is formed at the ninth or tenth cell cycle, when the cells at the vegetal edge of the blastoderm fuse with the underlying yolk cell.

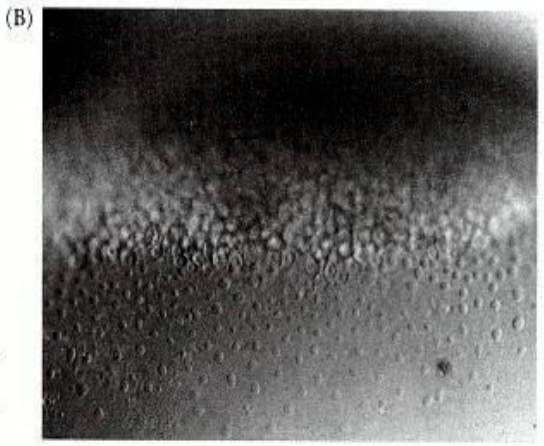
# Cleavage in Fish

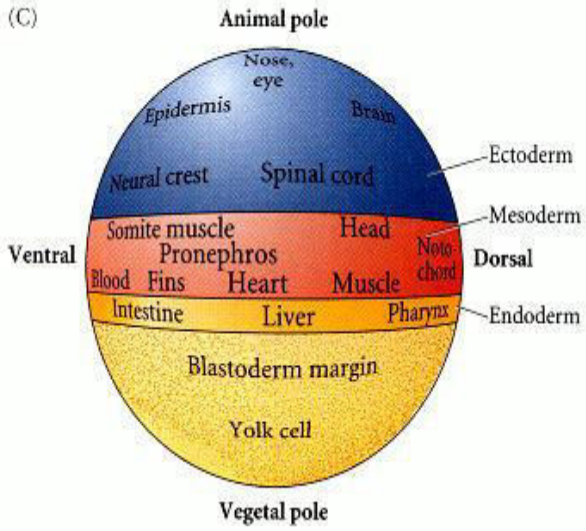
Later, as the blastoderm expands vegetally to surround the yolk cell, some of the yolk syncytial nuclei will move under the blastoderm to form the internal YSL, and some of the nuclei will move vegetally, staying ahead of the blastoderm margin, to form external YSL.

# Cleavage in Fish

The second cell population distinguished at the midblastula transition is the **enveloping layer (EVL).**



# Cleavage in Fish

It is made up of the most superficial cells of the blastoderm, which form an epithelial sheet a single cell layer thick. The EVL eventually becomes the **periderm,** an extra-embryonic protective covering that is sloughed off during later development.

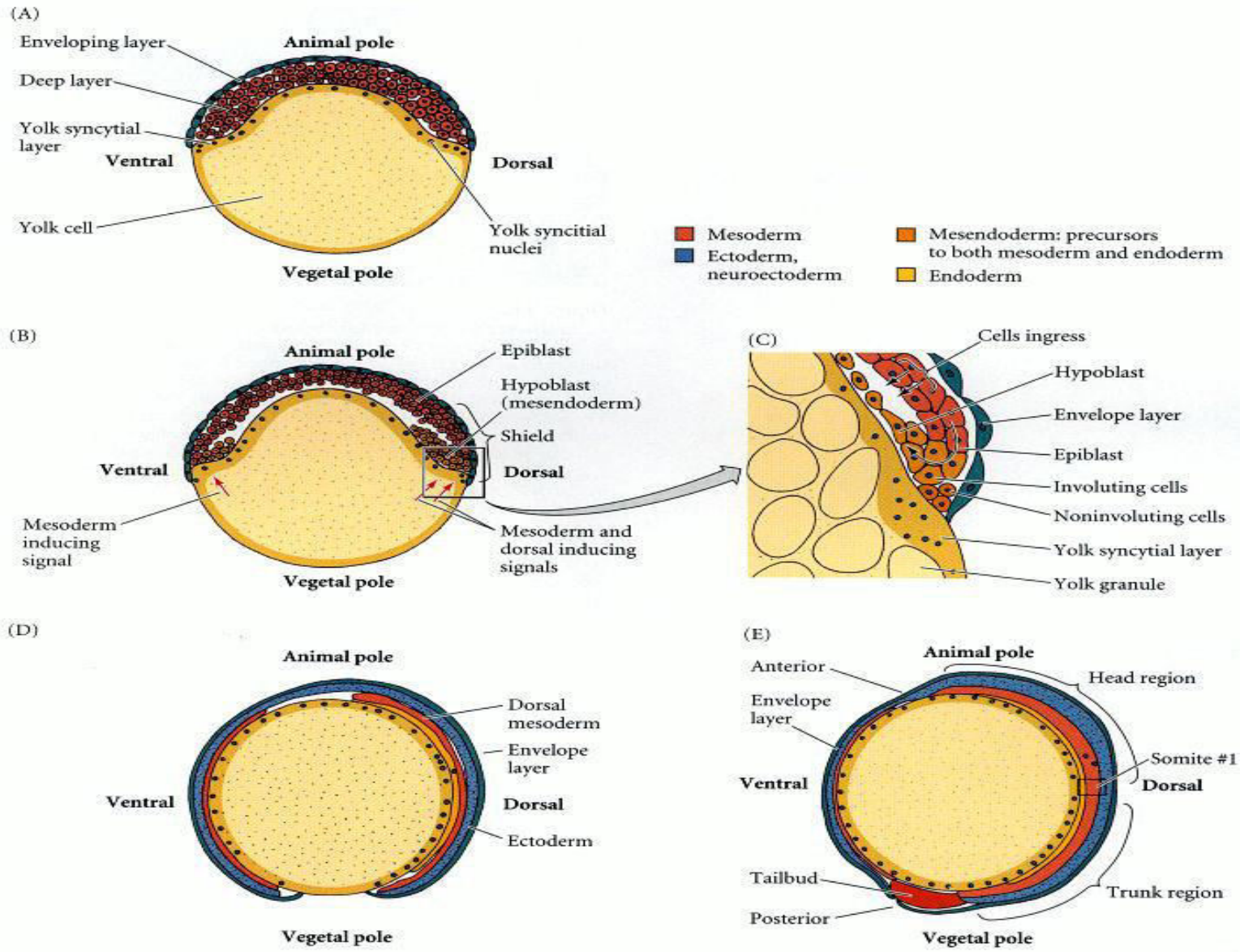
# Cleavage in Fish

Between the EVL and the YSL are the deep cells. These are the cells that give rise to the embryo proper. The fates of the early blastoderm cells are not determined, and cell lineage studies show that there is much cell mixing during cleavage.

# Early Gastrulation in Fish Embryos

# Gastrulation in Fish Embryos

The first cell movement of fish gastrulation is the epiboly of the blastoderm cells over the yolk.



# Gastrulation in Fish Embryos

This movement is not due to the active crawling of the blastomeres, however. Rather, the movement is provided by the autonomously expanding YSL "within" the animal pole yolk cytoplasm.

The EVL is tightly joined to the YSL and is dragged along with it.

# Gastrulation in Fish Embryos

The deep cells of the blastoderm then fill in the space between the **YSL** and the **EVL** as epiboly proceeds.

This can be demonstrated by severing the attachments between the **YSL** and the **EVL.**

# Layers formation during Gastrulation in Fish Embryos

# Gastrulation in Fish Embryos

When this is done, the **EVL** and deep cells spring back to the top of the yolk, while the **YSL** continues its expansion around the yolk cell. The expansion of the YSL depends on a network of microtubules in the **YSL**, and radiation or drugs that block the polymerization of tubulin inhibit epiboly

# Gastrulation in Fish Embryos

During migration, one side of the blastoderm becomes noticeably thicker than the other. Cell-labelling experiments indicate that the thicker side marks the site of the future dorsal surface of the embryo

# Gastrulation in Fish Embryos

After the blastoderm cells have covered about half the zebra fish yolk cell, a thickening occurs throughout the margin of the epibolizing blastoderm. This thickening is called the **germ ring**, and it is composed of a superficial layer, the **epiblast**, and an inner layer, the **hypoblast**.

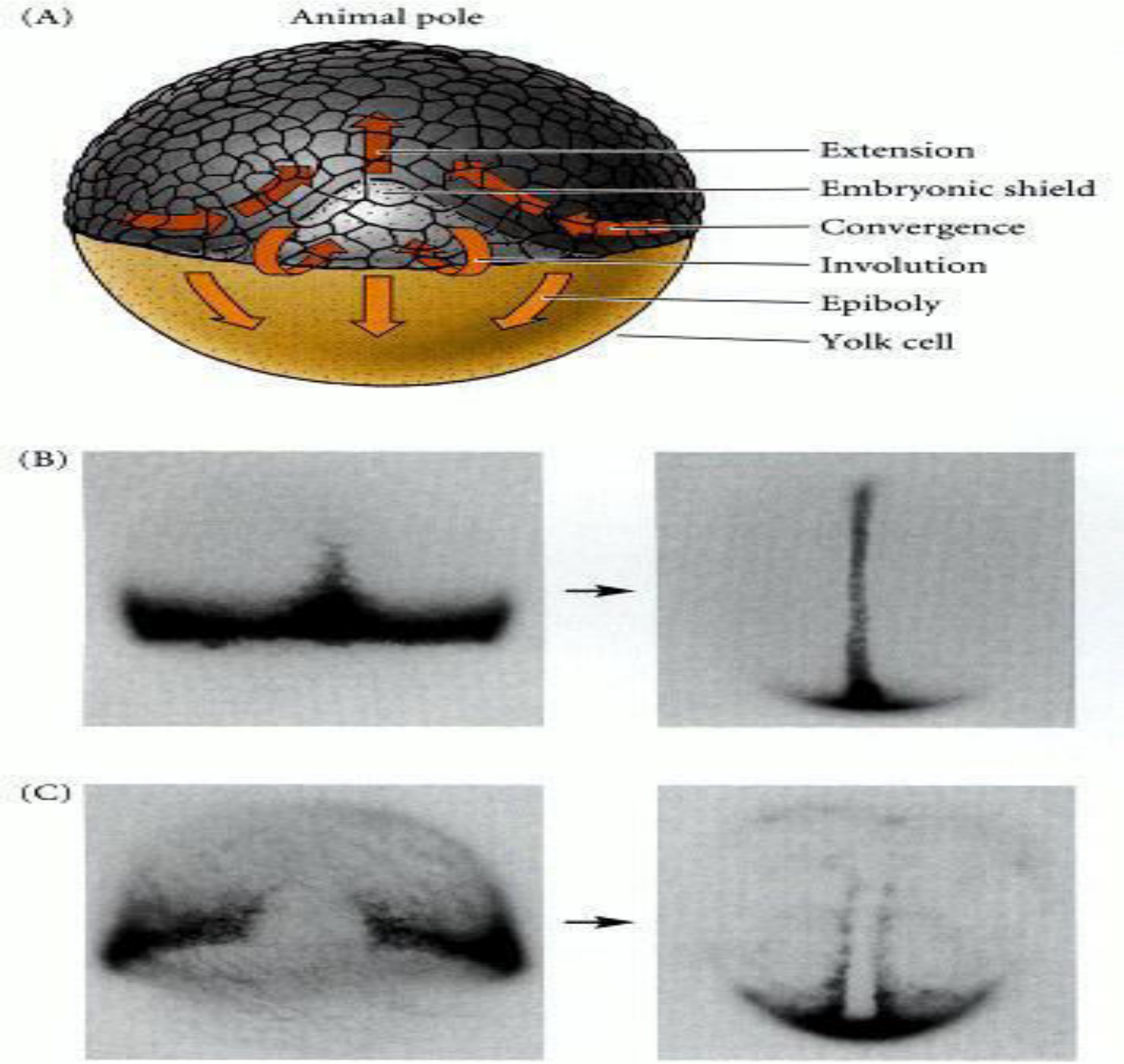
# Gastrulation in Fish Embryos II

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# Gastrulation in Fish Embryos II

The cells of both the epiblast and hypoblast intercalate on the future dorsal side of the embryo to form a localized thickening, the **embryonic shield**.



# Gastrulation in Fish Embryos II

The hypoblast cells of the embryonic shield converge and extend anteriorly, eventually narrowing along the dorsal midline of the hypoblast. This movement forms the **chorda mesoderm**, the precursor of the notochord.

# Gastrulation in Fish Embryos II

The cells adjacent to the chordamesoderm, the **paraxial mesoderm** cells, are the precursors of the mesodermal somites.

# Gastrulation in Fish Embryos II

The concomitant convergence and extension in the epiblast brings the presumptive neural cells from all over the epiblast into the dorsal midline, where they form the neural keel. The rest of the epiblast becomes the skin of the fish.

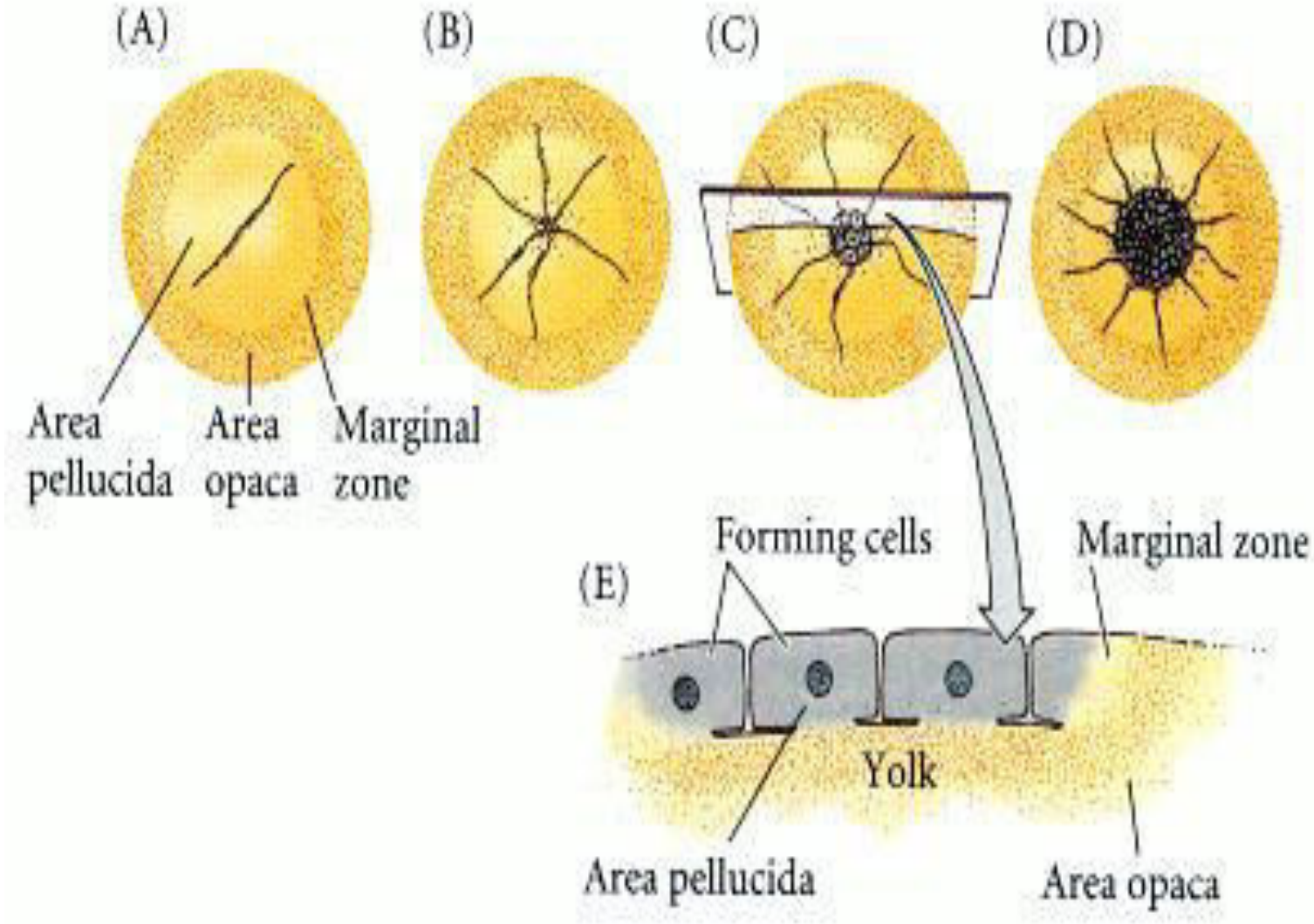
# Cleavage in Bird Eggs

# Cleavage in Bird Eggs

Fertilization of the chick egg occurs in the oviduct, before the albumen and the shell are secreted upon it. The egg is telo-lecithal (like that of the fish),with a small disc of cytoplasm sitting atop a large yolk.

# Cleavage in Bird Eggs

Cleavage occurs only in the blastodisc, a small disc of cytoplasm 2-3 mm in diameter at the animal pole of the egg cell. The first cleavage furrow appears centrally in the blastodisc, and other cleavages follow to create a single-layered blastoderm.



# Cleavage in Bird Eggs II

# Cleavage in Bird Eggs

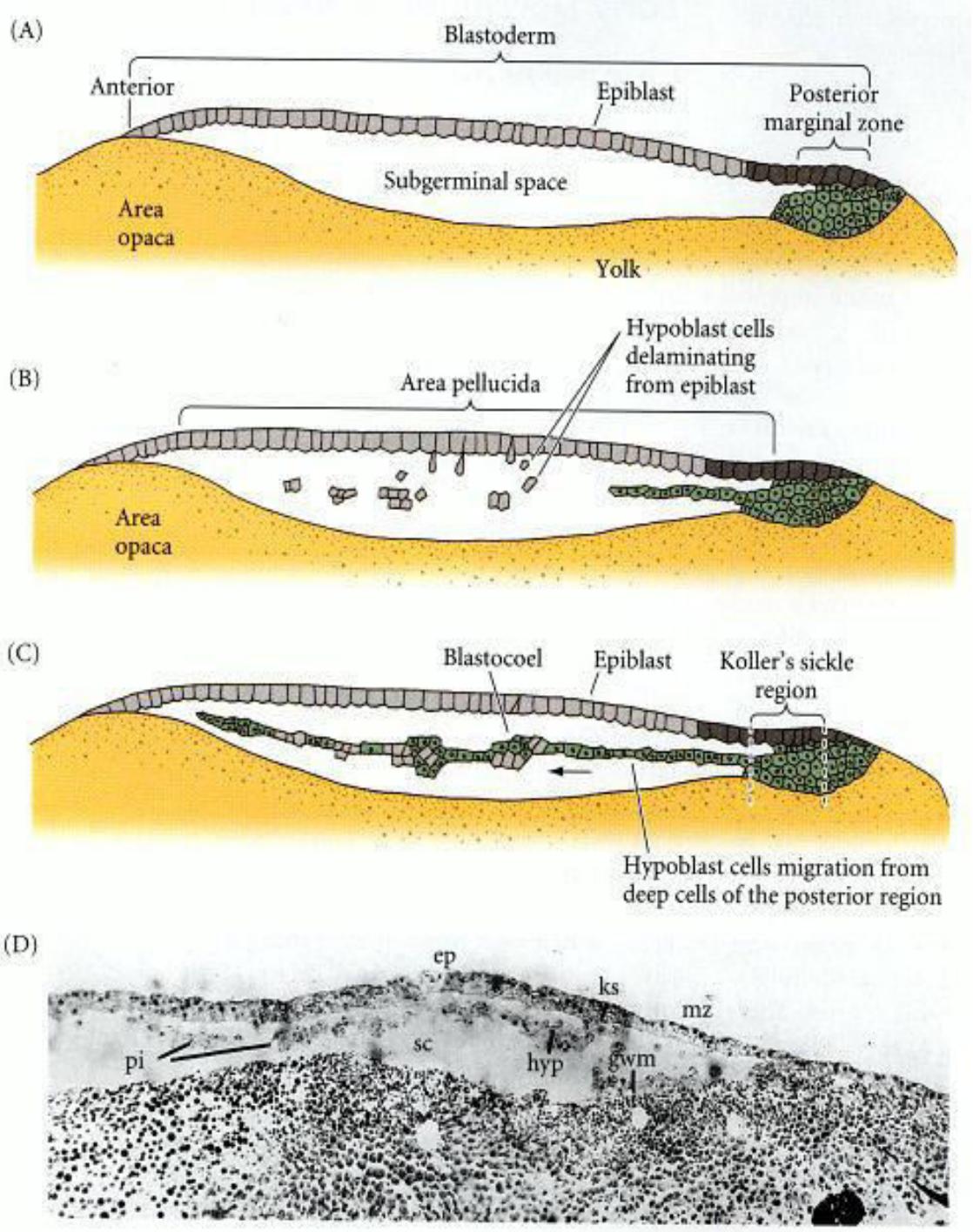
Between the blastoderm and the yolk is a space called the sub-germinal cavity. This space is created when the blastoderm cells absorb fluid from the albumin ("egg white") and secrete it between themselves and the yolk.

# Cleavage in Bird Eggs

At this stage, the deep cells in the center of the blastoderm are shed and die, leaving behind a onecell- thick area pellucida. This part of the blastoderm forms most of the actual embryo. The peripheral ring of blastoderm cells that have not shed their deep cells constitutes the **area opaca**.

# Cleavage in Bird Eggs

Between the area pellucida and the area opaca is a thin layer of cells called the **marginal zone** (or **marginal belt**).

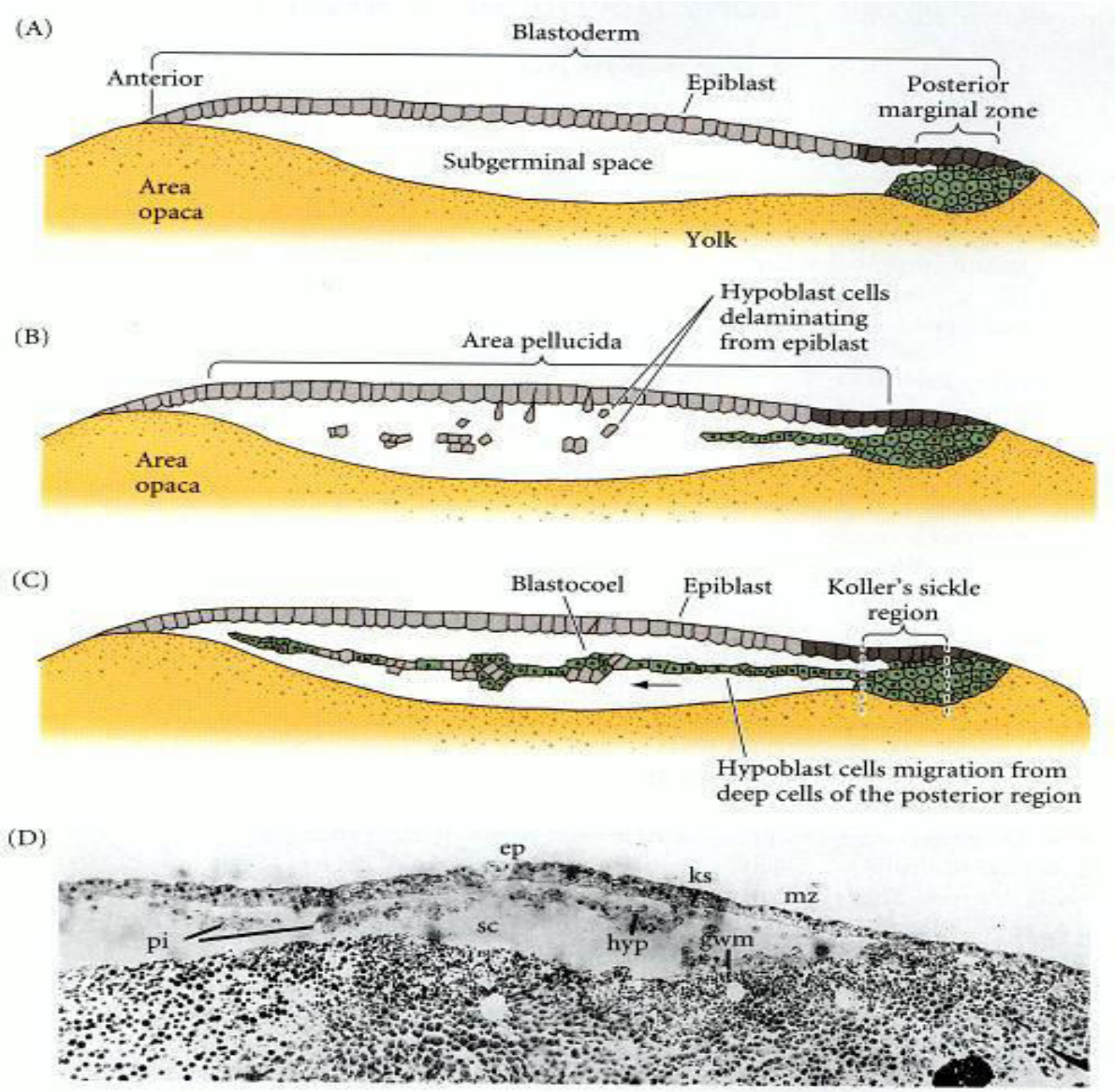


# Gastrulation of the Avian Embryo

# Gastrulation of the Avian Embryo

## The hypoblast

Area pellucida cells migrate individually into the sub-germinal cavity to form the poly-invagination islands (primary hypoblast).

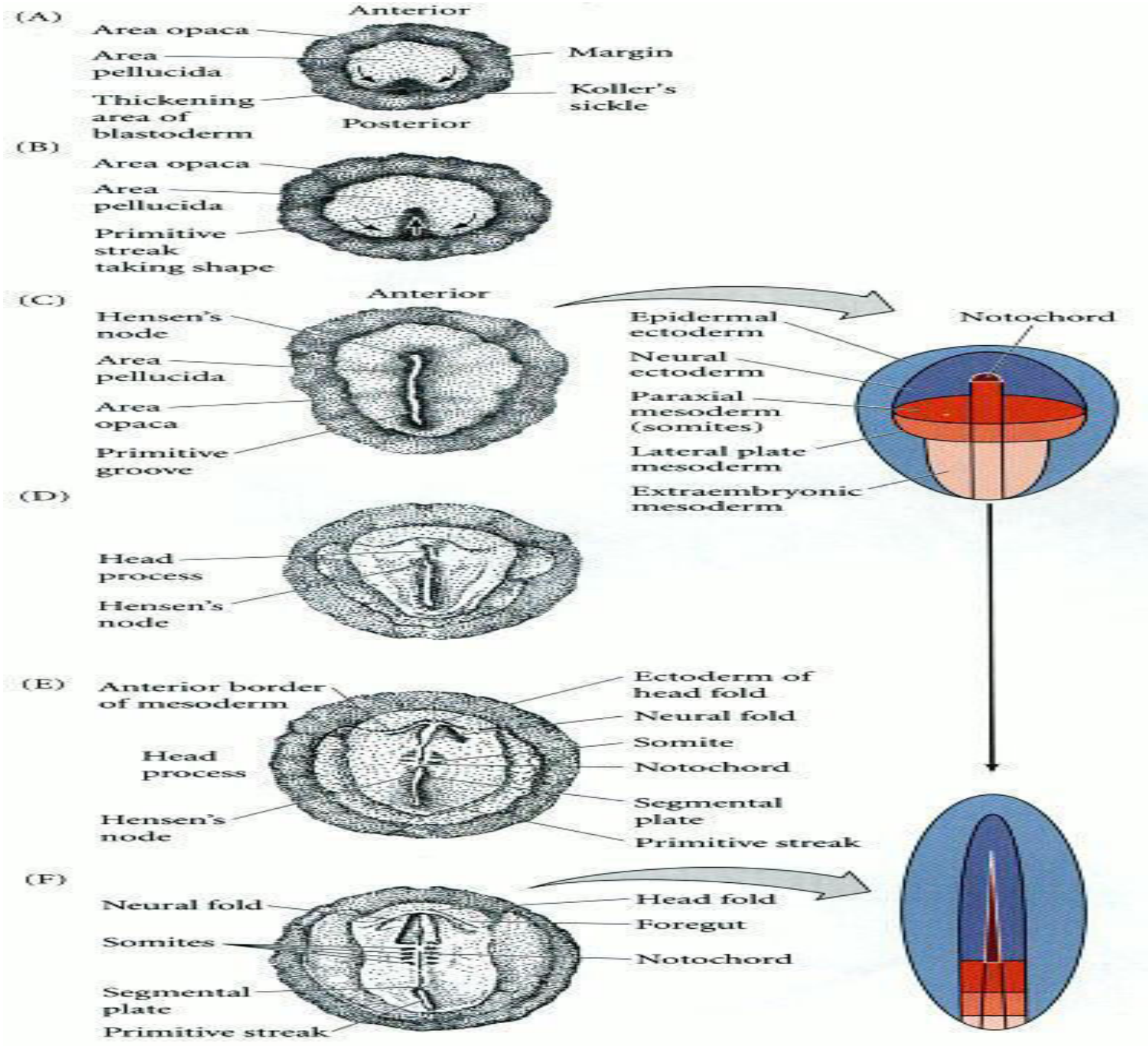


# Gastrulation of the Avian Embryo

Shortly thereafter, a sheet of cells from the *posterior* margin of the blastoderm (distinguished from the other regions of the margin by Koller's sickle, a local thickening) migrates anteriorly to join the poly-invagination islands, thereby forming the secondary hypoblast.

# Gastrulation of the Avian Embryo

The two-layered blastoderm (epiblast and hypoblast) is joined together at the margin of the area opaca, and the space between the layers forms a blastocoel.



# Gastrulation of the Avian Embryo II

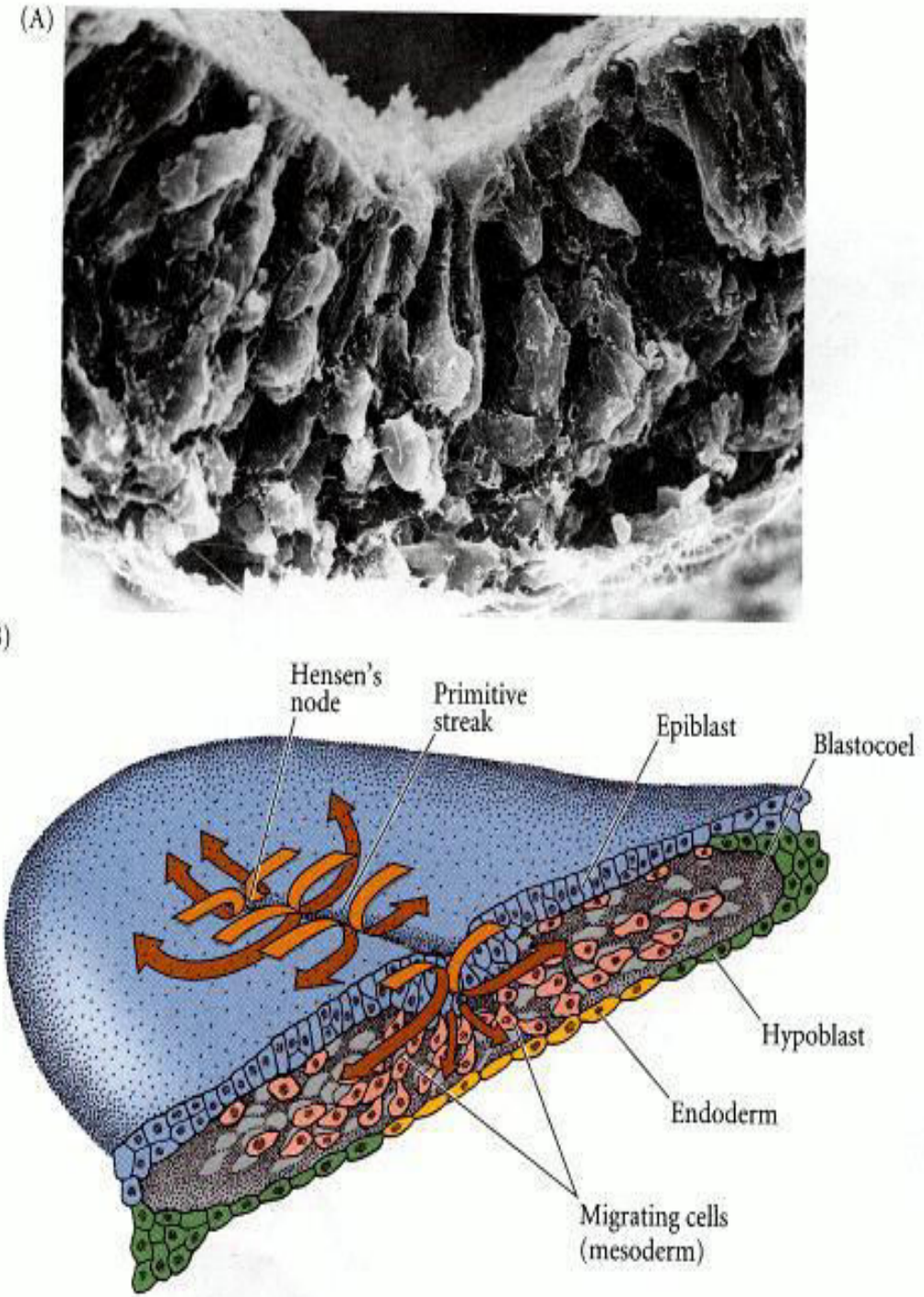
# Gastrulation of the Avian Embryo

## The primitive streak

The major structural characteristic of avian, reptilian, and mammalian gastrulation is the **primitive streak**. This streak is first visible as a thickening of the epiblast at the posterior region of the embryo, just anterior to Koller's sickle

# Gastrulation of the Avian Embryo

As cells converge to form the primitive streak, a depression forms within the streak. This depression is called the **primitive groove**, and it serves as an opening through which migrating cells pass into the blastocoel.



# Gastrulation of the Avian Embryo

At the anterior end of the primitive streak is a regional thickening of cells called the **primitive knot** or **Hensen's node**. The center of this node contains a funnel-shaped depression (sometimes called the **primitive pit**) through which cells can pass into the blastocoel.

# Gastrulation of the Avian Embryo

Cells migrating through **Hensen's node** pass down into the blastocoel and migrate anteriorly, forming foregut, head mesoderm, and notochord; cells passing through the lateral portions of the primitive streak give rise to the majority of endodermal and mesodermal tissues.

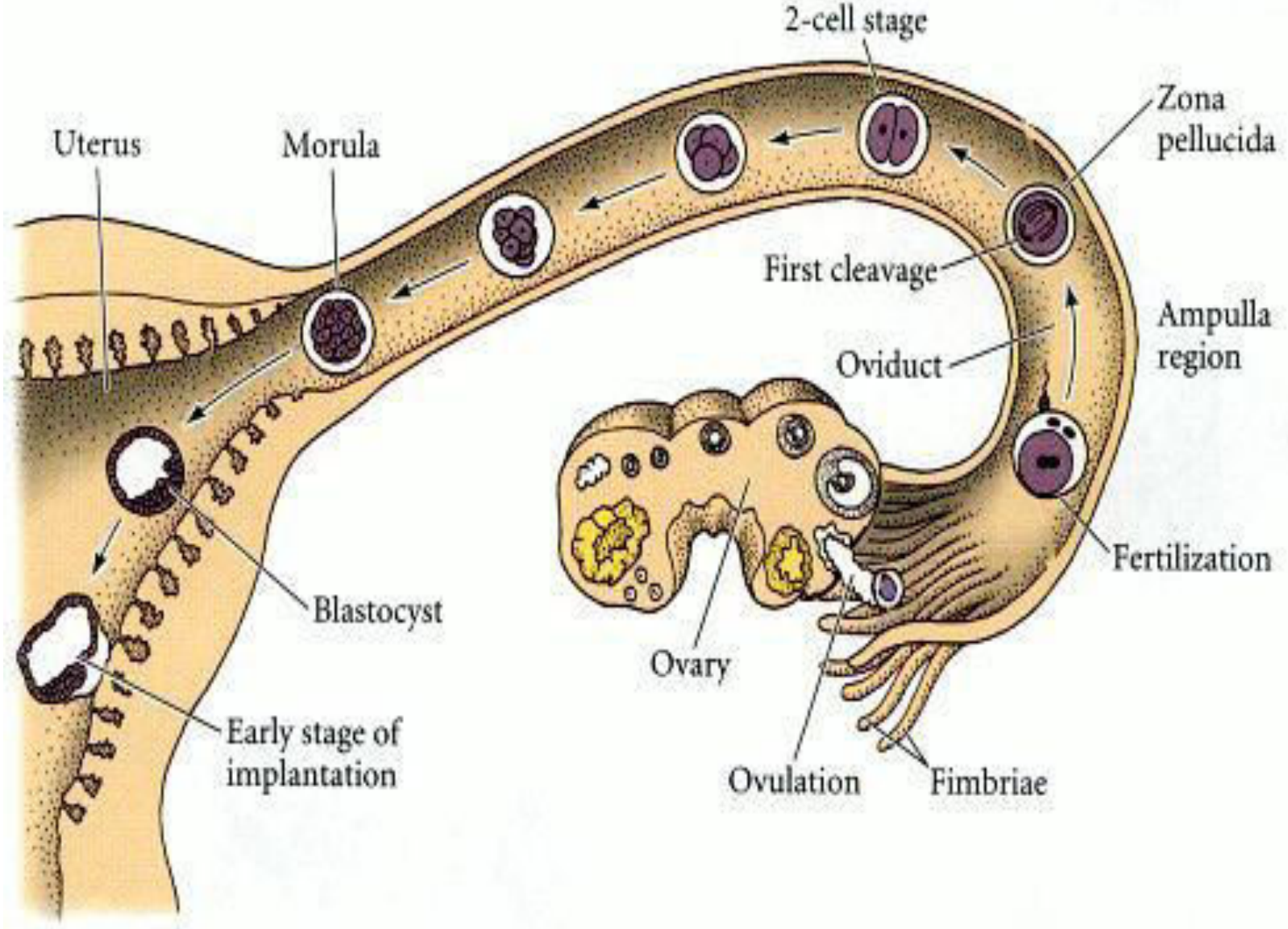
# Cleavage in Mammals

# Cleavage in Mammals

The mammalian oocyte is released from the ovary and swept by the fimbriae into the oviduct. Fertilization occurs in the ampulla of the oviduct, a region close to the ovary. Meiosis is completed at this time, and first cleavage begins about a day later.

# Cleavage in Mammals

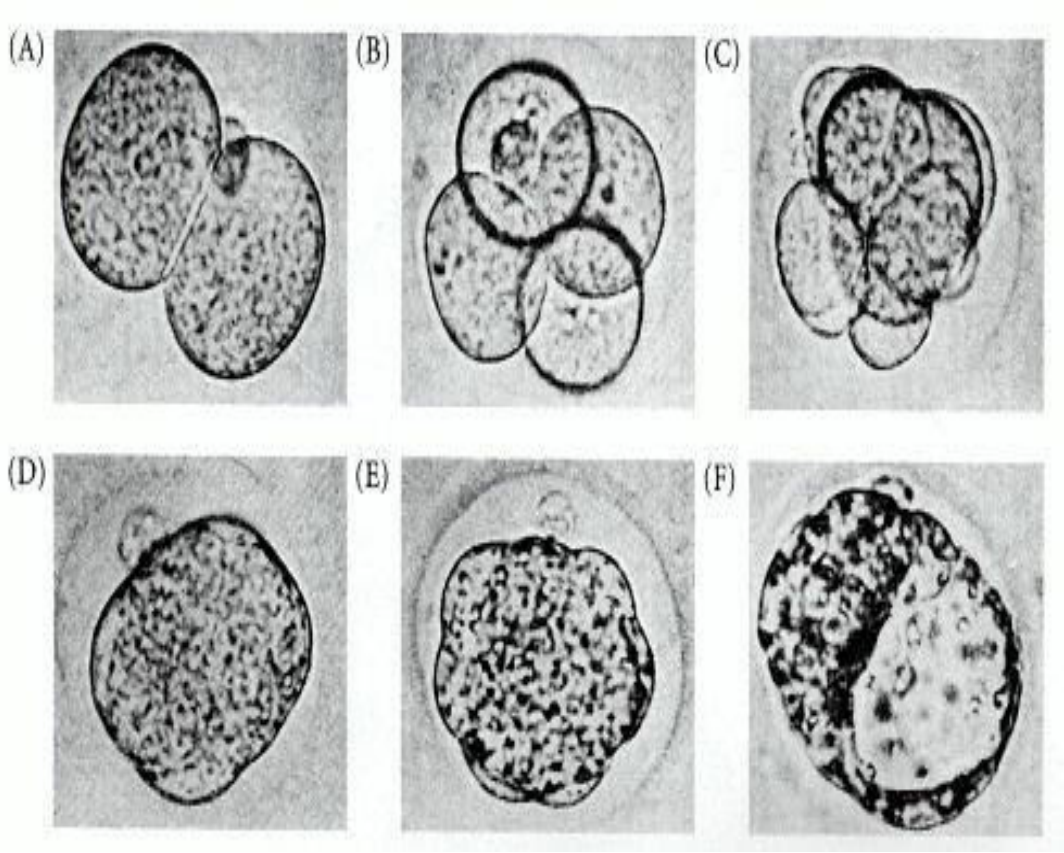
Cleavages in mammalian eggs are among the slowest in the animal kingdom about 12-24 hours apart. Meanwhile, the cilia in the oviduct push the embryo toward the uterus; the first cleavages occur along this journey.



# Cleavage in Mammals II

# Cleavage in Mammals

The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of **compaction**.



# Cleavage in Mammals

The cells of the compacted 8-cell embryo divide to produce a 16-cell **morula**. The morula consists of a small group of internal cells surrounded by a larger group of external cells.

# Cleavage in Mammals

Most of the descendants of the external cells become the trophoblast (trophectoderm) cells. This group of cells produces no embryonic structures. Rather, it forms the tissue of the chorion, the embryonic portion of the placenta.

# Cleavage in Mammals III

# Cleavage in Mammals

These cells generate the **inner cell mass (ICM),** which will give rise to the embryo and its associated yolk sac, allantois, and amnion. By the 64-cell stage, the inner cell mass (approximately 13 cells) and the trophoblast cells have become separate cell layers, neither contributing cells to the other group.

# Cleavage in Mammals

During a process called **cavitation**, the trophoblast cells secrete fluid into the morula to create a blastocoel. The inner cell mass is positioned on one side of the ring of trophoblast cells. The resulting structure, called the **blastocyst**, is another hallmark of mammalian cleavage.

# Cleavage in Mammals

Once out, the blastocyst can make direct contact with the uterus. The uterine epithelium **(endometrium)** "catches" the blastocyst on an extracellular matrix containing collagen, laminin, fibronectin, hyaluronic acid, and heparin sulfate receptors.

# Gastrulation in Mammals

# Gastrulation in Mammals

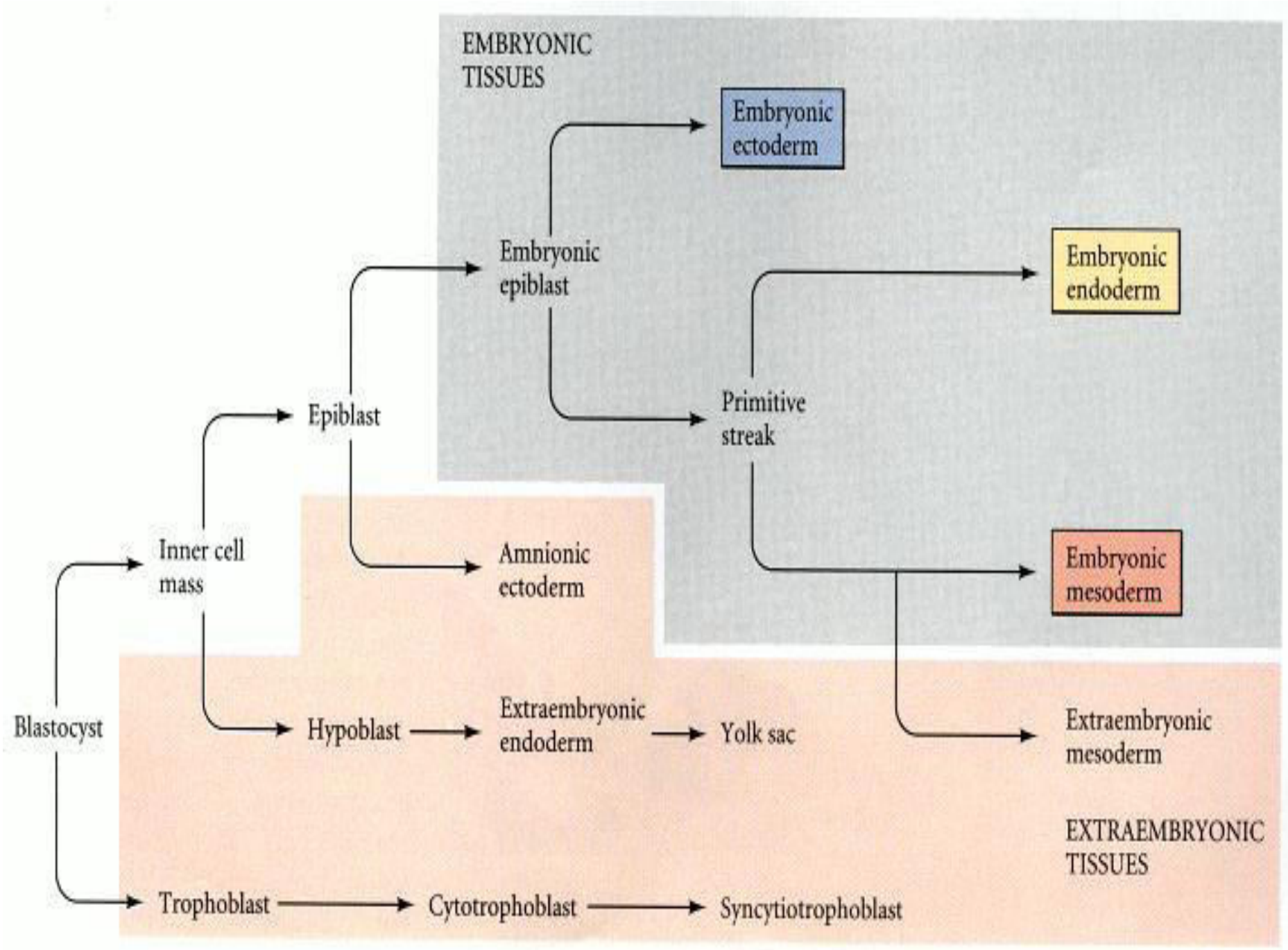
The mammalian embryo obtains nutrients directly from its mother and does not rely on stored yolk. This adaptation has entailed a dramatic restructuring of the maternal anatomy (such as expansion of the oviduct to form the uterus) as well as the development of a fetal organ capable of absorbing maternal nutrients.

# Gastrulation in Mammals

This fetal organ the **Chorion** is derived primarily from embryonic trophoblast cells, supplemented with mesodermal cells derived from the inner cell mass. The chorion forms the fetal portion of the placenta.

# Gastrulation in Mammals

It will induce the uterine cells to form the maternal portion of the placenta, the **decidua**. The decidua becomes rich in the blood vessels that will provide oxygen and nutrients to the embryo.



# Gastrulation in Mammals II

# Gastrulation in Mammals

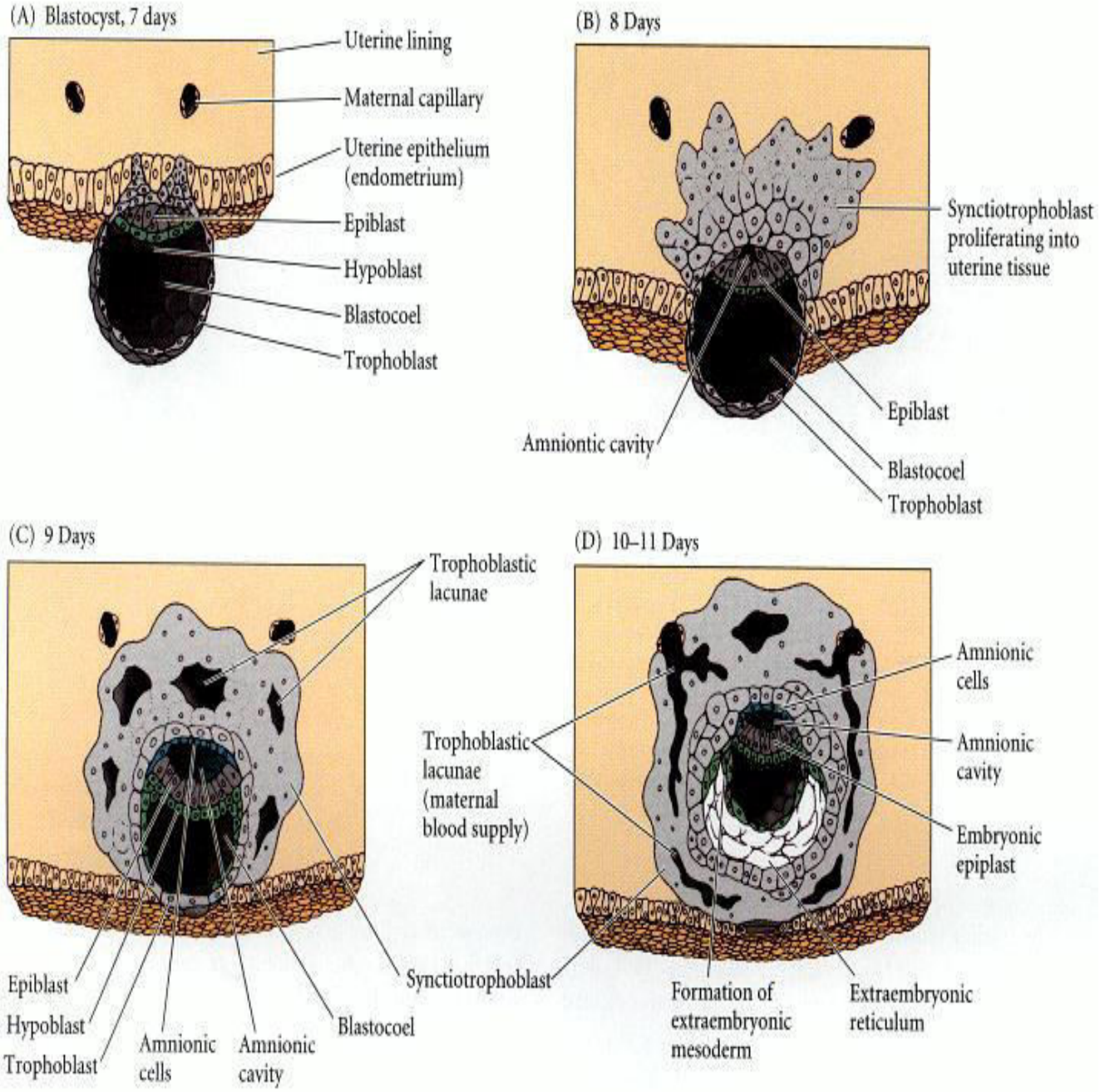
The first segregation of cells within the inner cell mass results in the formation of the hypoblast (sometimes called the primitive endoderm) layer. The hypoblast cells delaminate from the inner cell mass to line the blastocoel cavity, where they give rise to the extra-embryonic endoderm, which forms the yolk sac.

# Gastrulation in Mammals

The epiblast cell layer is split by small clefts that eventually coalesce to separate the embryonic epiblast from the other epiblast cells, which form the **amnionic cavity**. Once the lining of the amnion is completed, it fills with a secretion called **amnionic (amniotic) fluid**, which serves as a shock absorber for the developing embryo while preventing its desiccation.

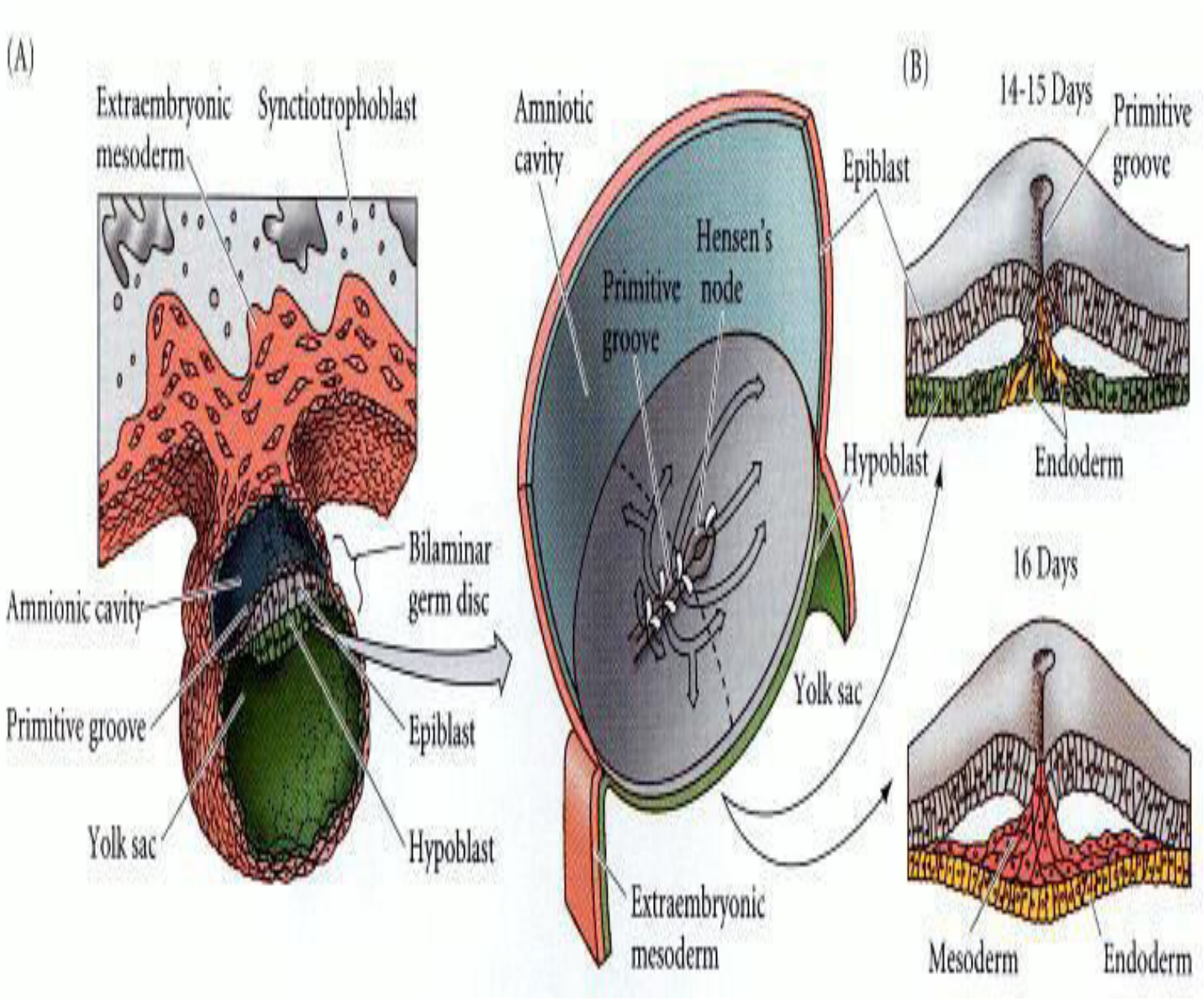
# Gastrulation in Mammals

Gastrulation begins at the posterior end of the embryo, and this is where the **node** forms.



# Gastrulation in Mammals

Those cells migrating through the node give



# Gastrulation in Mammals III

# Gastrulation in Mammals II

The original type of trophoblast cells constitute a layer called the cytotrophoblast, whereas the multinucleated type of cell forms the syncytiotrophoblast. The cytotrophoblast initially adheres to the endometrium through a series of adhesion molecules.

# Gastrulation in Mammals II

The extra-embryonic mesoderm joins the trophoblastic extensions and gives rise to the blood vessels that carry nutrients from the mother to the embryo. The narrow connecting stalk of extra-embryonic mesoderm that links the embryo to the trophoblast eventually forms the vessels of the **umbilical cord**.

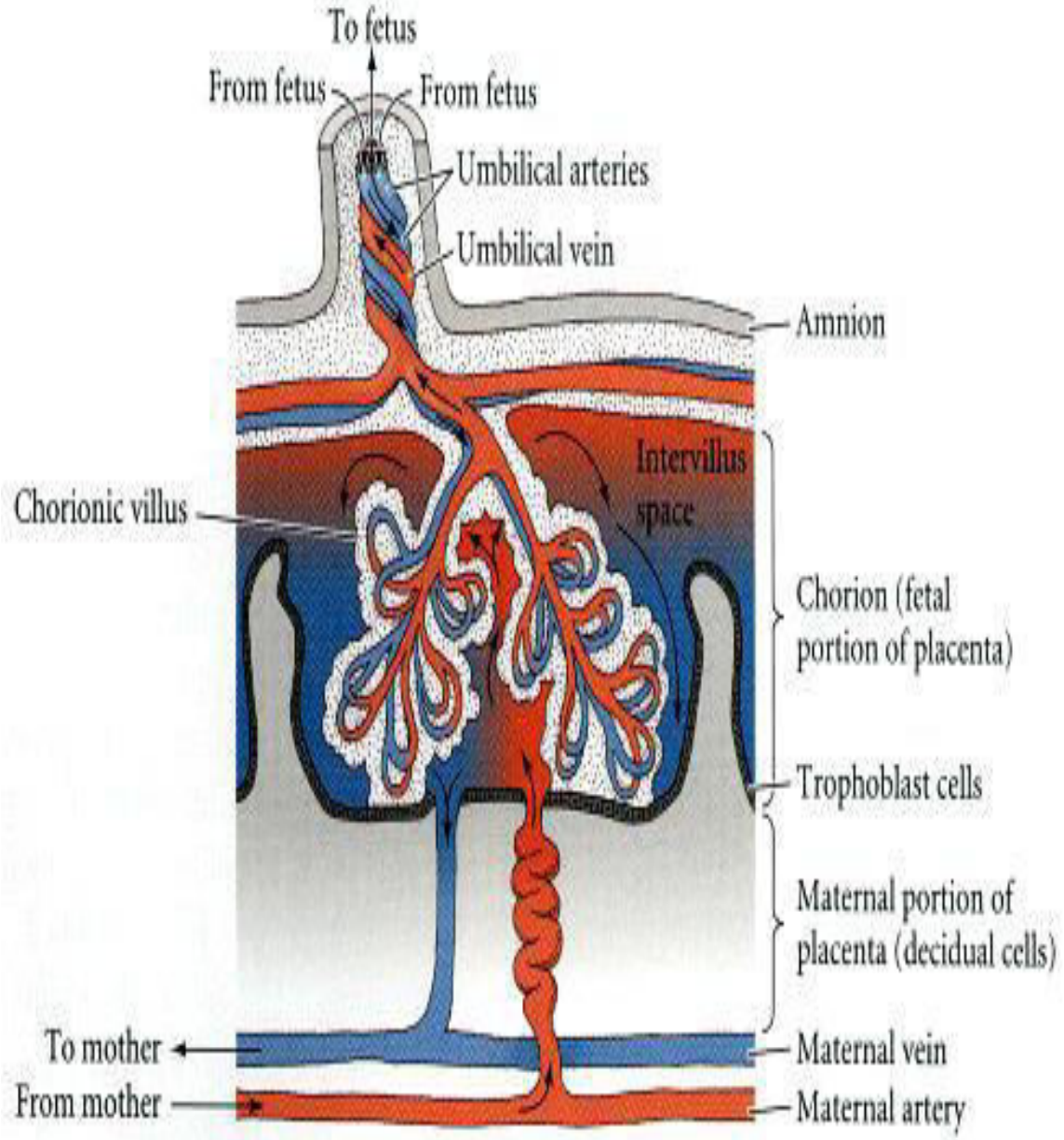
# Gastrulation in Mammals II

The fully developed extra-embryonic organ, consisting of trophoblast tissue and blood vessel-containing mesoderm, is called the chorion, and it fuses with the uterine wall to create the placenta. Thus, the placenta has both a maternal portion (the uterine endometrium, which is modified during pregnancy) and a fetal component (the chorion).

# Gastrulation in Mammals IV

# Gastrulation in Mammals II

Figure shows the relationships between the embryonic and extra-embryonic tissues of a 6-week human embryo. The embryo is seen encased in the amnion and is further shielded by the chorion.



# Gastrulation in Mammals IV

The blood vessels extending to and from the chorion are readily observable, as are the villi that project from the outer surface of the chorion. These villi contain the blood vessels and allow the chorion to have a large area exposed to the maternal blood.

# Gastrulation in Mammals IV

Mother provides the fetus with nutrients and oxygen, and the fetus sends its waste products (mainly carbon dioxide and urea) into the maternal circulation.

The maternal and fetal blood cells, however, usually do not mix.

# Summary
# The Early Development of Vertebrates

# The Early Development of Vertebrates

- Fishes, reptiles, and birds undergo discoidal meroblastic cleavage, wherein the early cell divisions do not cut through the yolk of the egg. These cells form a blastoderm.

# The Early Development of Vertebrates

- In fishes, the deep cells form between the yolk syncytial layer and the enveloping layer. These cells migrate over the top of the yolk, forming the hypoblast and epiblast layers.

# The Early Development of Vertebrates

- There appear to be two signaling centers supplying anterior-posterior information in fishes, one located at the border between the neural and surface ectoderm, the other in the lateral mesoderm.

# Summary the Early Development of Vertebrates II

# The Early Development of Vertebrates

- In chick embryos, early cleavage forms an area opaca and an area pellucida. The region between them is the marginal zone. Gastrulation begins at the posterior marginal zone, as the hypoblast and primitive streak both start there.

# The Early Development of Vertebrates

- The primitive streak is derived from anterior epiblast cells and the central cells of the posterior marginal zone. As the primitive streak extends rostrally, Hensen's node is formed. Cells migrating through Hensen's node become chordamesoderm (notochord) cells. These extend up to the presumptive midbrain, where they meet the prechordal plate.

# The Early Development of Vertebrates

- The **prechordal plate** induces the formation of the forebrain; the **chordamesoderm** induces the formation of the midbrain, hindbrain, and spinal cord. The first cells migrating laterally through the primitive streak become endoderm, displacing the hypoblast.

# Summary The Early Development of Vertebrates III

# The Early Development of Vertebrates III

- In birds, gravity is critical in determining the anterior-posterior axis, while pH differences appear crucial for distinguishing dorsal from ventral. The left-right axis is formed by the expression of *nodal* on the left side of the embryo.

# The Early Development of Vertebrates III

- Mammals undergo holoblastic rotational cleavage, characterized by a slow rate of division, a unique cleavage orientation, lack of divisional synchrony, and the formation of a blastocyst.

# The Early Development of Vertebrates III

- The blastocyst forms after the blastomeres undergo compaction. It contains outer cells the trophoblast cells that become the chorion, and an inner cell mass that becomes the amnion and the embryo.

# The Early Development of Vertebrates III

- The chorion forms the fetal portion of the placenta, which functions to provide oxygen and nutrition to the embryo, to provide hormones for the maintenance of pregnancy, and to provide barriers to the mother's immune system.

# Summary the Early Development of Vertebrates IV

# The Early Development of Vertebrates IV

- Mammalian gastrulation is not unlike that of birds. There appear to be two signaling centers one in the node and one in the anterior visceral endoderm. The latter is critical for generating the forebrain, while the former is critical in inducing the axial structures caudally from the midbrain.

# The Early Development of Vertebrates IV

- Hox genes pattern the anterior-posterior axis and help to specify positions along that axis. If Hox genes are knocked out, segment-specific malformations can arise. Similarly, causing the ectopic expression of Hox genes can alter the body axis.

# The Early Development of Vertebrates IV

- The homology of gene structure and the similarity of expression patterns between *Drosophila* and mammalian **Hox genes** suggests that this patterning mechanism is extremely ancient.

# Development of Twins

# Development of Twins

Each ICM (Inner Cell Mass) cell has the same potency (in this case, each cell can give rise to all the cell types of the embryo, but not to the trophoblast), and their fates will be determined by interactions among their descendants.

# Development of Twins

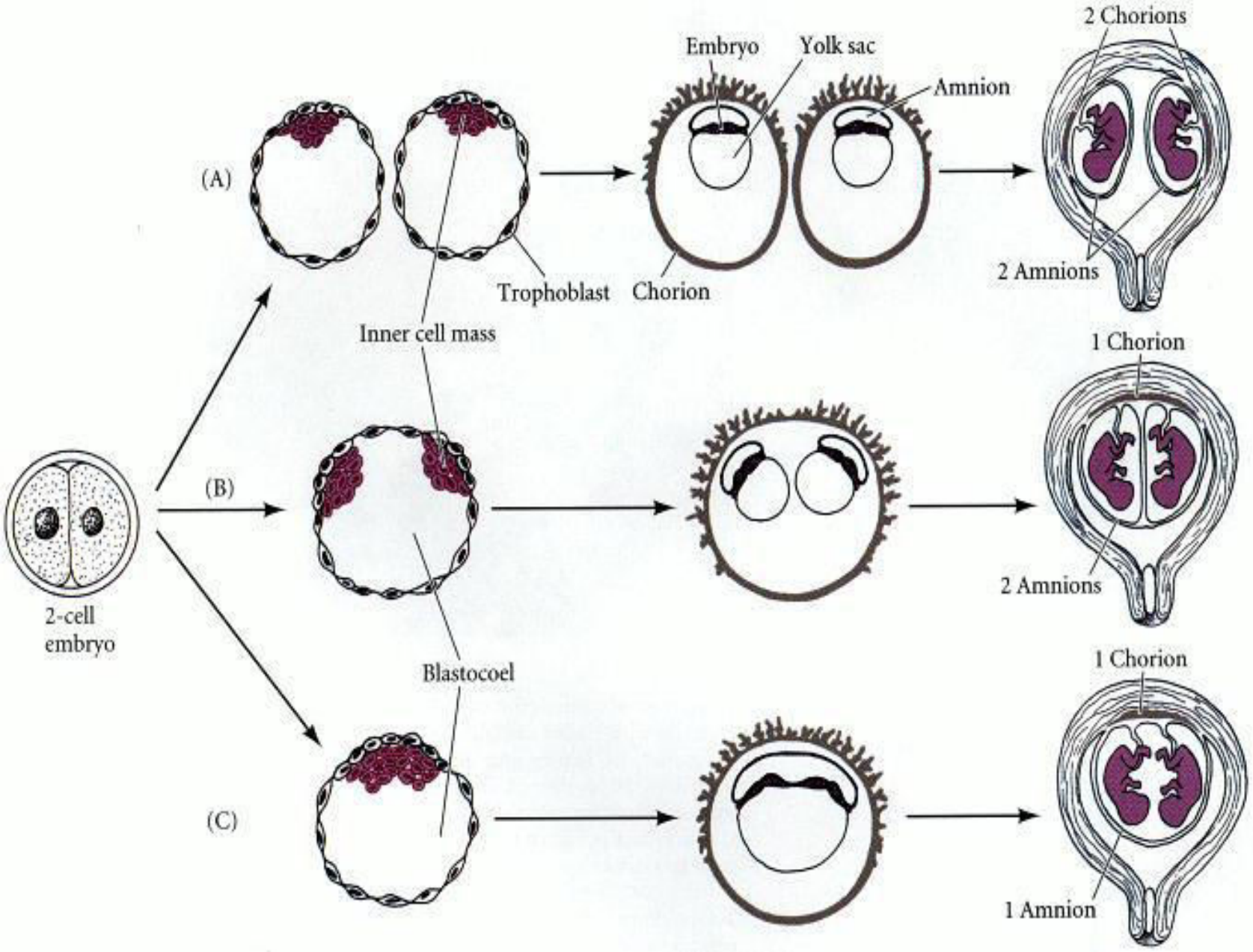
The regulative capacity of the ICM blastomeres is also seen in humans. Human twins are classified into two major groups: **monozygotic** (one-egg, or identical) twins and **dizygotic** (two egg, or fraternal) twins.

# Development of Twins

Fraternal twins are the result of two separate fertilization events, whereas identical twins are formed from a single embryo whose cells somehow dissociated from one another.

# Development of Twins

Identical twins occur in roughly **0.25%** of human births. About **33%** of identical twins have two complete and separate chorions, indicating that separation occurred before the formation of the trophoblast tissue at **day 5.**



# Development of Twins

# Development of Twins

The remaining identical twins share a common chorion, suggesting that the split occurred within the inner cell mass after the trophoblast formed.

- By day 9, the human embryo has completed the construction of another extraembryonic layer, the **lining of the amnion.**

# Development of Twins

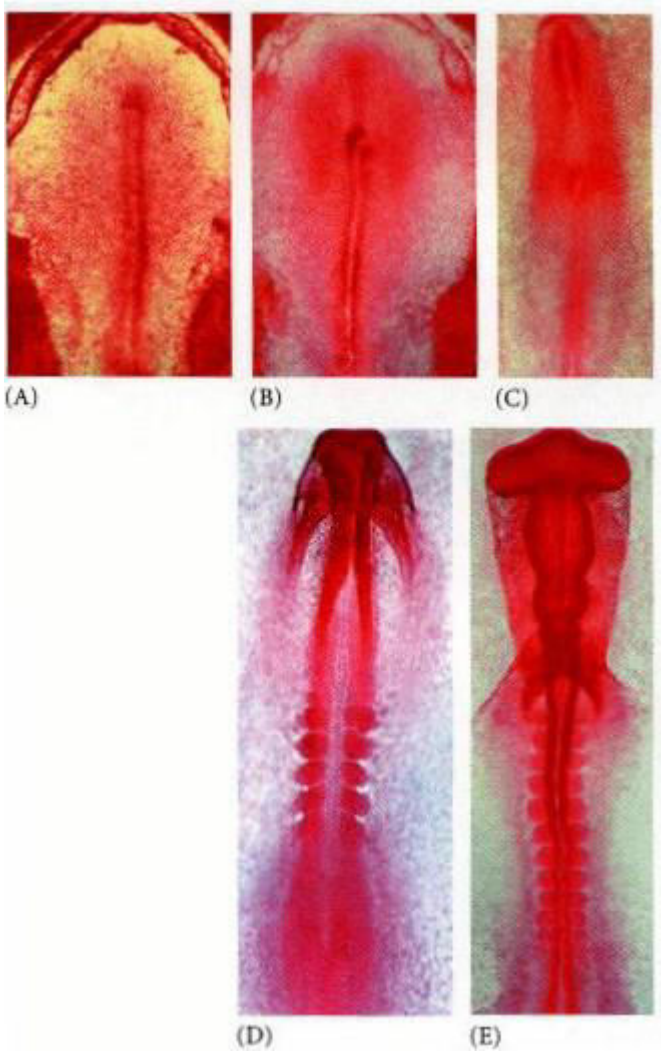
This tissue forms the amnionic sac (or water sac), which surrounds the embryo with amnionic fluid and protects it from desiccation and abrupt movement.

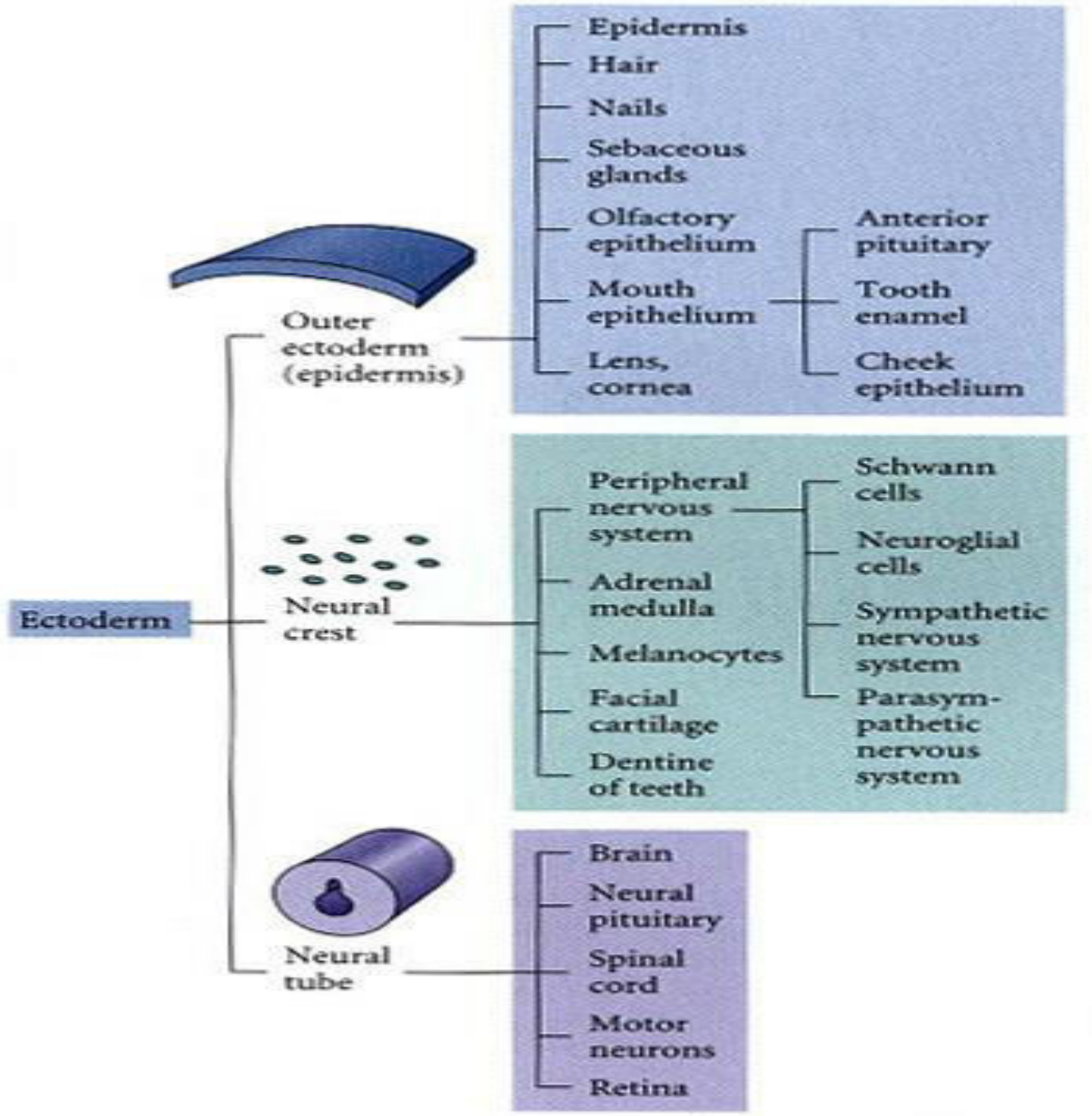
- If the separation of the embryo were to come after the formation of the chorion on day 5 but before the formation of the amnion on day 9, then the resulting embryos should have one chorion and two amnions.

# Development of central nervous system

# Development of central nervous system

The fates of the vertebrate ectoderm is shown in figure.



# Development of central nervous system

A portion of the dorsal ectoderm is specified to become neural ectoderm, and its cells become distinguishable by their columnar appearance. This region of the embryo is called the **neural plate**. The process by which this tissue forms a **neural tube**.

# Development of central nervous system

The rudiment of the central nervous system, is called **neurulation**, and an embryo undergoing such changes is called a **neurula**. The neural tube will form the brain anteriorly and the spinal cord.

# Formation of the Neural Tube

# Development of central nervous system

## Formation of the Neural Tube

There are two major ways of forming a neural tube

- **Primary neurulation**
- **Secondary neurulation**

# Development of central nervous system

In primary neurulation, the cells surrounding the neural plate direct the neural plate cells to proliferate, invaginate, and pinch off from the surface to form a hollow tube.

# Development of central nervous system

In secondary neurulation, the neural tube arises from a solid cord of cells that sinks into the embryo and subsequently hollows out to form a hollow tube. Neurulation in fishes is exclusively secondary.

# Development of central nervous system

In amphibians, such as *Xenopus,* most of the tadpole neural tube is made by primary neurulation, but the tail neural tube is derived from secondary neurulation.

In mice (and probably humans, too), secondary neurulation begins at or around the level of somite 35.

# Events in Primary Neurulation

# Events in Primary Neurulation

During primary neurulation, the original ectoderm is divided into three sets of cells.

- The internally positioned neural tube, which will form the brain and spinal cord.
- The externally positioned epidermis of the skin.

# Events in Primary Neurulation

- The neural crest cells. The neural crest cells form in the region that connects the neural tube and epidermis, but then migrate elsewhere; they will generate the peripheral neurons and glia, the pigment cells of the skin, and several other cell types.

# Formation of neural groove during Primary Neurulation

# Formation of neural groove during Primary Neurulation

Shortly after the neural plate has formed, its edges thicken and move upward to form the **neural folds**, while a U-shaped neural groove appears in the centre of the plate, dividing the future right and left sides of the embryo.

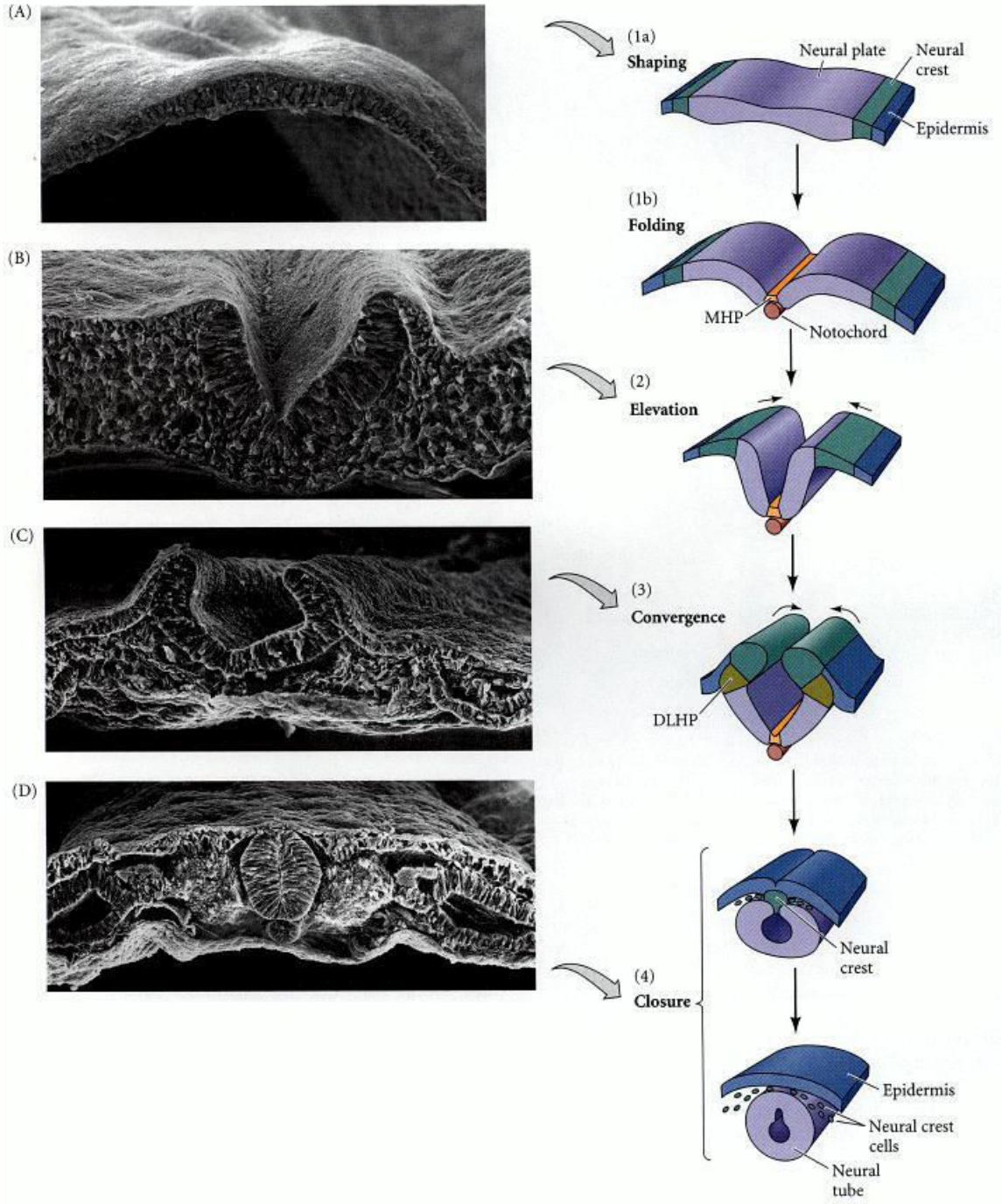
# Formation of neural groove during Primary Neurulation

The neural folds migrate toward the midline of the embryo, eventually fusing to form the neural tube beneath the overlying ectoderm. The cells at the dorsal most portion of the neural tube become the **neural crest** cells.

# Primary Neurulation in chick and Mammals

# Primary neurulation

The events of primary neurulation in the chick are illustrated in Figure

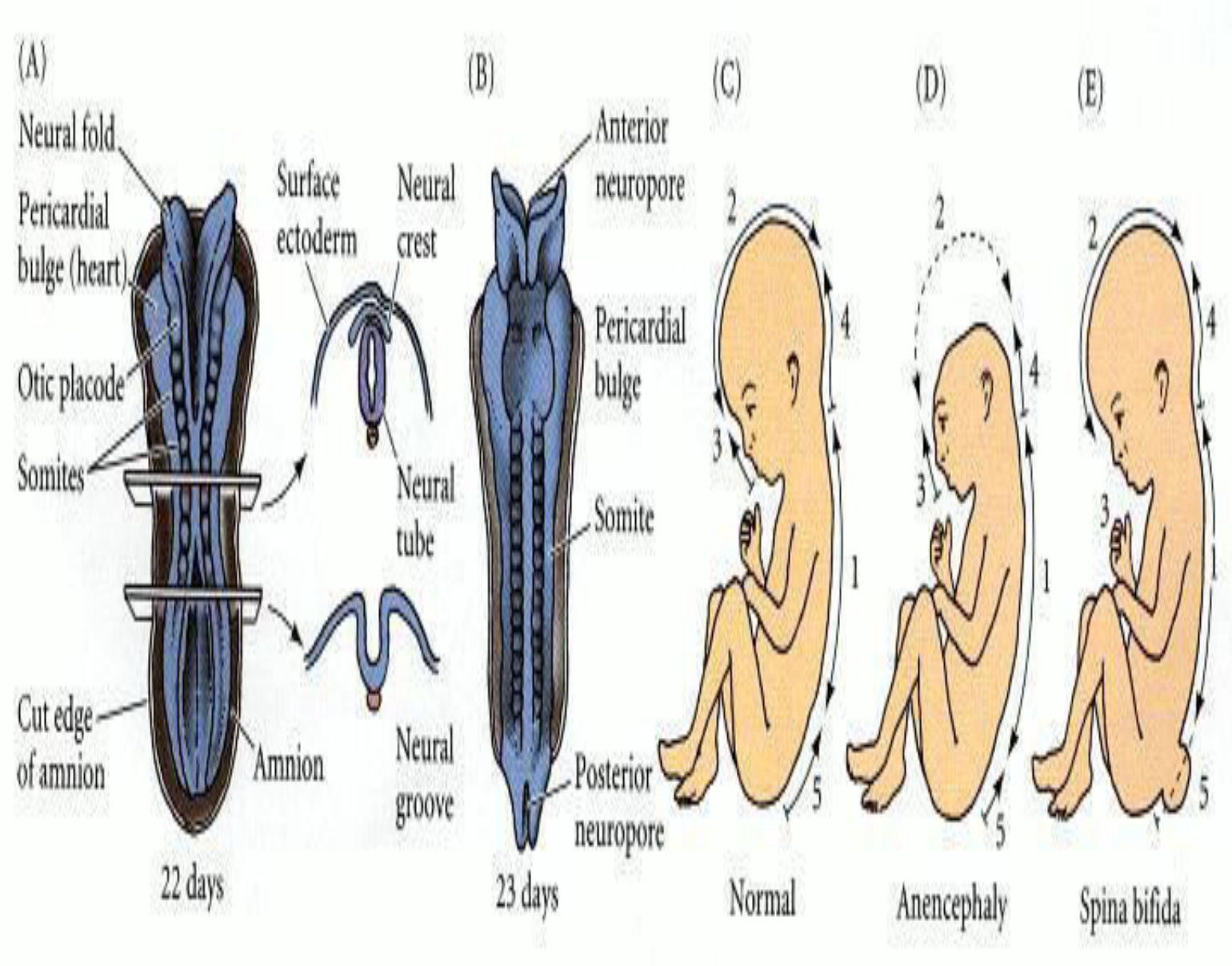


# Primary neurulation

In birds and mammals, the cells at the midline of the neural plate are called the **medial hinge point (MHP) cells**. They are derived from the portion of the neural plate just anterior to Hensen's node and from the anterior midline of Hensen's node.

# Primary neurulation

Different neural tube defects are caused when various parts of the neural tube fail to close.



# Abnormalities in Primary neurulation

# Primary neurulation

Failure to close the human *posterior* neural tube regions at day 27 (or the subsequent rupture of the posterior neuropore shortly thereafter) results in a condition called **spina bifida**, the severity of which depends on how much of the spinal cord remains exposed.

# Primary neurulation

Failure to close the *anterior* neural tube regions results in a lethal condition, **anencephaly**. Here, the forebrain remains in contact with the amniotic fluid and subsequently degenerates.

Fetal forebrain development ceases, and the vault of the skull fails to form.

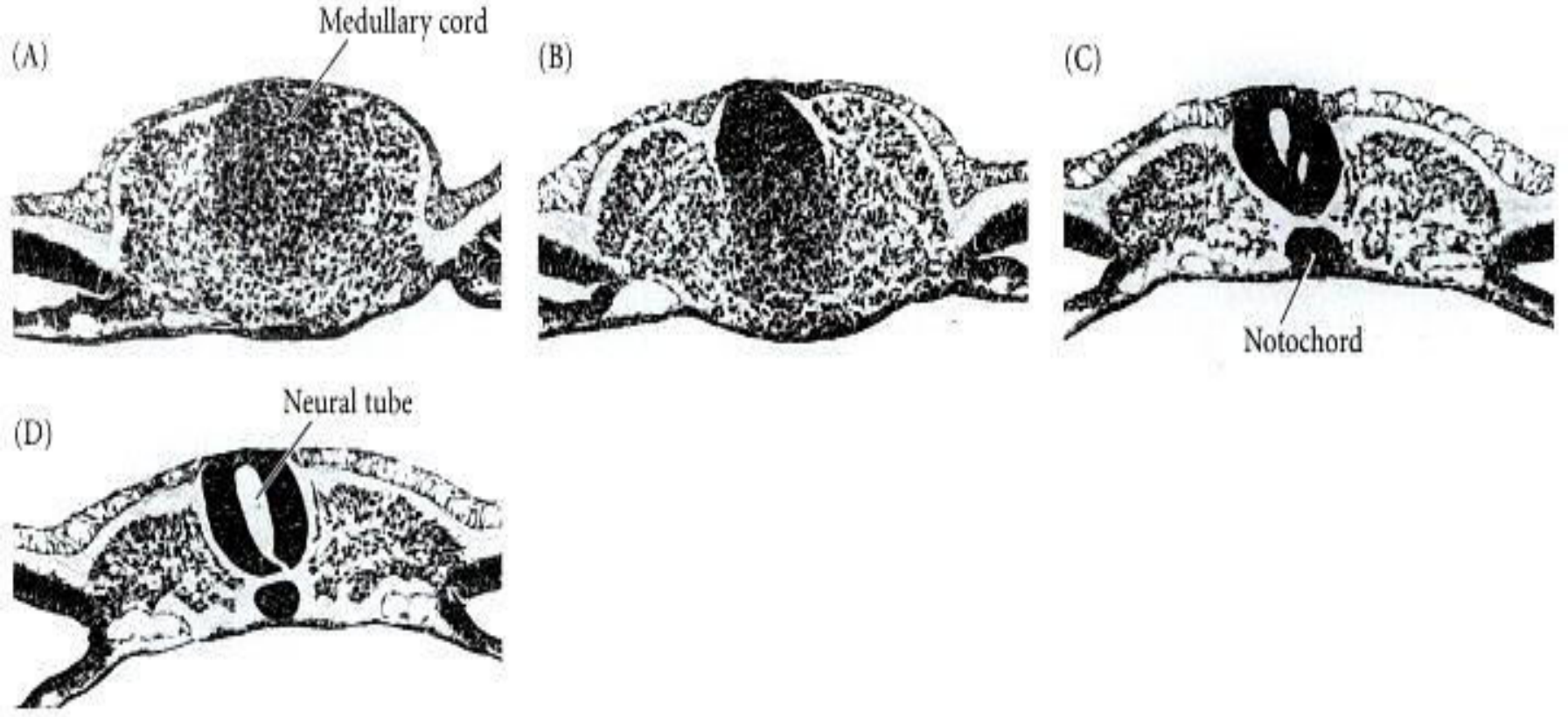
# Primary neurulation

The failure of the entire neural tube to close over the entire body axis is called Craniorachischisis. Collectively, neural tube defects are not rare in humans, as they are seen in about 1 in every 500 live births.

# Secondary neurulation

# Secondary neurulation

Secondary neurulation involves the making of a **medullary cord** and its subsequent hollowing into a neural tube.



# Secondary neurulation

In frogs and chicks, secondary neurulation is usually seen in the neural tube of the lumbar (abdominal) and tail vertebrae. In both cases, it can be seen as a continuation of gastrulation.

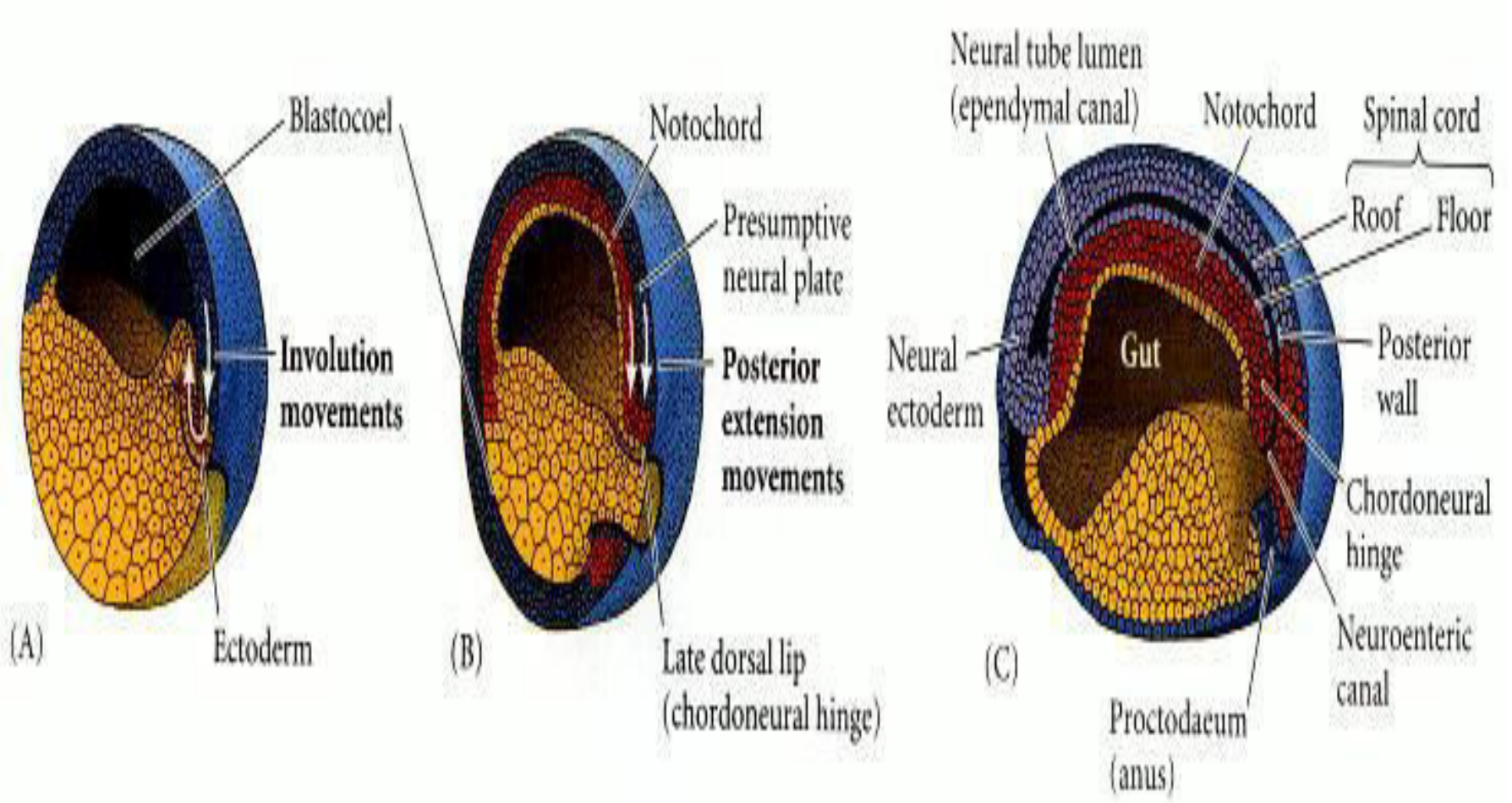
# Secondary neurulation II

# Secondary neurulation

The growing region at the tip of the lip is called the chordoneural hinge, and it contains precursors for both the posterior most portion of the neural plate and the posterior portion of the notochord.

# Secondary neurulation

The tip of the tail is the direct descendant of the dorsal blastopore lip, and the cells lining the



# Secondary neurulation

The proximal part of the neurenteric canal fuses with the anus, while the distal portion becomes the ependymal canal.

# Differentiation of the Neural Tube

# Differentiation of the Neural Tube

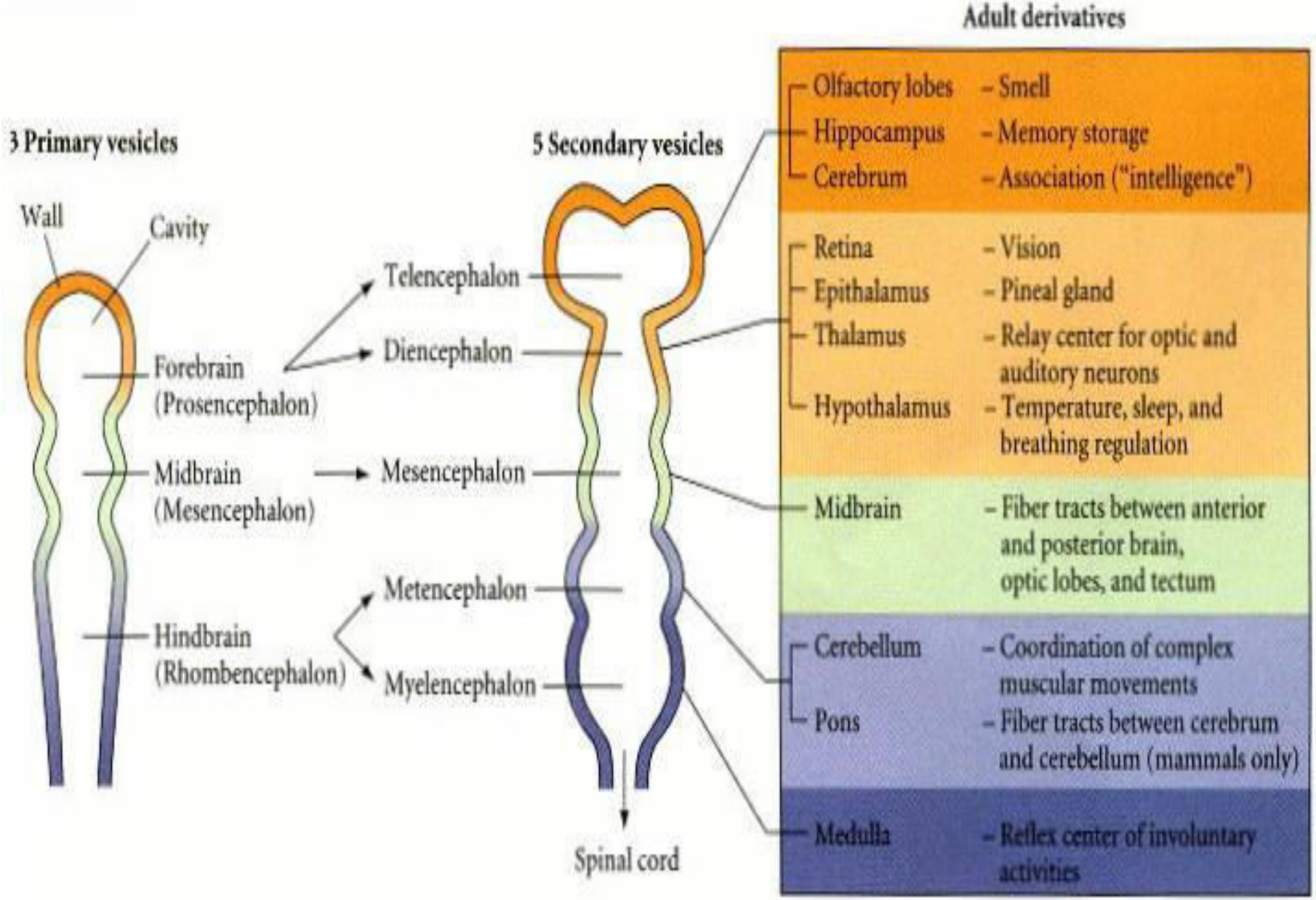
## The anterior-posterior axis

In this region, the neural tube balloons into three primary vesicles.

- Forebrain (prosencephalon),
- Midbrain (mesencephalon),
- Hindbrain (rhombencephalon).

# Differentiation of the Neural Tube

The Prosencephalon becomes subdivided into the anterior telencephalon and the more caudal diencephalon.



# Differentiation of the Neural Tube II

# Differentiation of the Neural Tube

The telencephalon will eventually form the **cerebral hemispheres**, and the diencephalon will form the **thalamic and hypothalamic brain regions** that receive neural input from the retina.

# Differentiation of the Neural Tube

The Rhombencephalon becomes subdivided into a posterior Myelencephalon and a more anterior Metencephalon. The Myelencephalon eventually becomes the Medulla oblongata, whose neurons generate the nerves that regulate respiratory, gastrointestinal, and cardiovascular movements

# Differentiation of the Neural Tube

The Metencephalon gives rise to the cerebellum, the part of the brain responsible for coordinating movements, posture, and balance.

# Differentiation of the Neural Tube III

# Differentiation of the Neural Tube III

## The dorsal-ventral axis

The neural tube is polarized along its dorsal-ventral axis. In the spinal cord, for instance, the *dorsal* region is the place where the spinal neurons receive input from sensory neurons, while the *ventral* region is where the motor neurons reside.

# Differentiation of the Neural Tube III

In the middle are numerous interneurons that relay information between them. The polarity of the neural tube is induced by signals coming from its immediate environment. The dorsal pattern is imposed by the epidermis, while the ventral pattern is induced by the notochord.

# Agent of ventral specification in neural tube differentiation

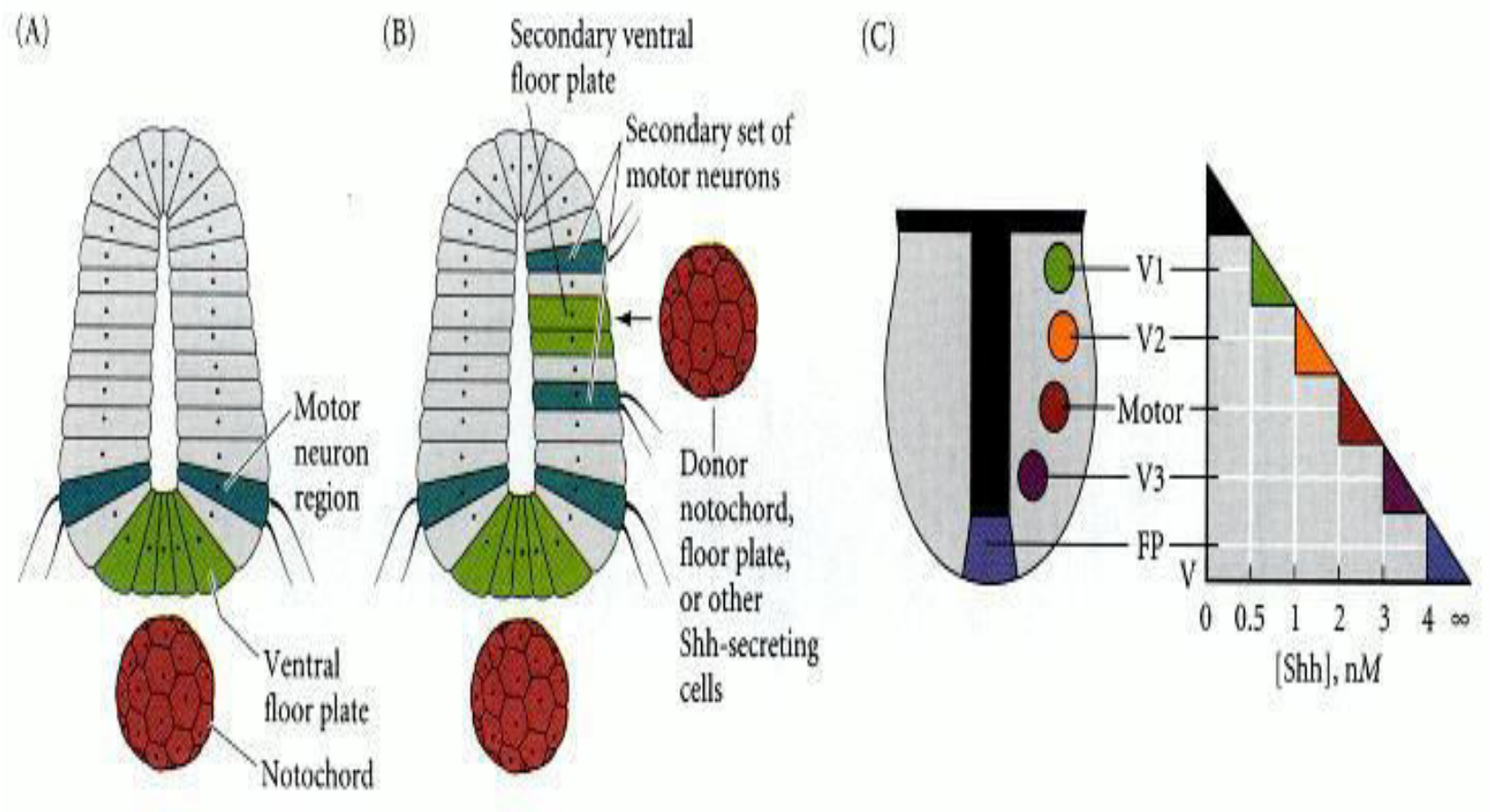
# Differentiation of the Neural Tube II

The specification of the ventral neural tube appears to be mediated by external tissues.

One agent of ventral specification is the Sonic hedgehog protein, probably originating from the notochord.

# Differentiation of the Neural Tube II

Another agent specifying the ventral neural cell types is **retinoic acid**, which probably comes from the adjacent somites.



# Differentiation of the Neural Tube II

The secreted Sonic hedgehog induces the medial hinge cells to become the **floor plate** of the neural tube. These floor plate cells also secrete Sonic hedgehog, which forms a gradient highest at the most ventral portion of the neural tube.

# Differentiation of the Neural Tube II

Those cells adjacent to the floor plate that receive high concentrations of Sonic hedgehog become the **ventral (V3) neurons**, while the next group of cells, exposed to slightly less Sonic hedgehog, become motor neurons.

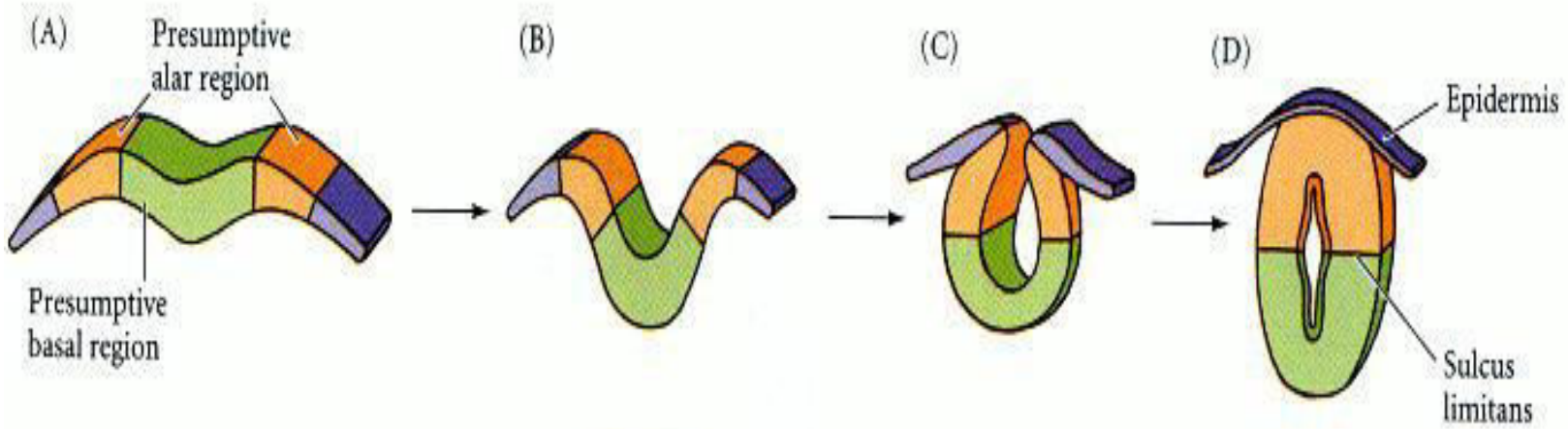
# Spinal chord and medulla organization

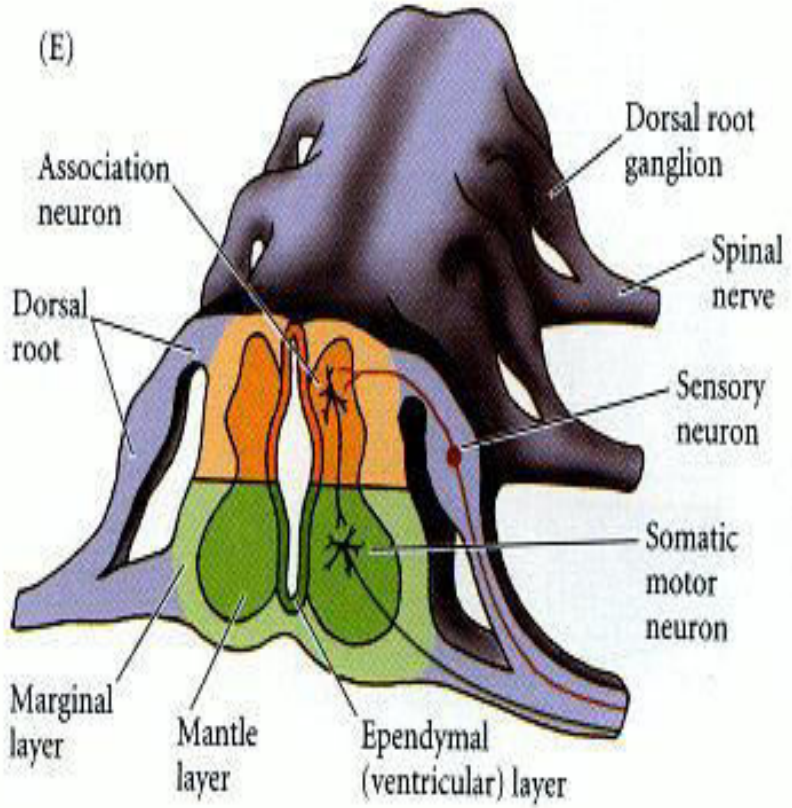
# Spinal chord and medulla organization

As the cells adjacent to the lumen continue to divide, the migrating cells form a second layer around the original neural tube. This layer becomes progressively thicker as more cells are added to it from the germinal neuroepithelium.

# Spinal chord and medulla organization

This new layer is called the **mantle** (or **intermediate**) **zone**, and the germinal epithelium is now called the **ventricular zone** (and, later, the **ependyma**).



# Spinal chord and medulla organization

The mantle zone cells differentiate into both neurons and glia.

The neurons make connections among themselves and send forth axons away from the lumen, thereby creating a cell-poor **marginal zone**.

# Spinal chord and medulla organization II

# Spinal chord and medulla organization

The mantle zone cells differentiate into both neurons and glia.

The neurons make connections among themselves and send forth axons away from the lumen, thereby creating a cell-poor **marginal zone**.

# Spinal chord and medulla organization

Eventually, glial cells cover many of the axons in the marginal zone in myelin sheaths, giving them a whitish appearance. Hence, the mantle zone, containing the neuronal cell bodies, is often referred to as the Gray matter; the axonal, marginal layer is often called the White matter.

# Spinal chord and medulla organization

In the spinal cord and medulla, this basic three-zone pattern of ependymal, mantle, and marginal layers is retained throughout development. The gray matter (mantle) gradually becomes a butterfly-shaped structure surrounded by white matter; and both become encased in connective tissue.

# Spinal chord and medulla organization

As the neural tube matures, a longitudinal groove the sulcus limitans divides it into dorsal and ventral halves. The dorsal portion receives input from sensory neurons, whereas the ventral portion is involved in effecting various motor functions.

# Cerebellar organization

# Cerebellar organization

In the cerebellum, some neuronal precursors enter the marginal zone to form clusters of neurons called nuclei. Each nucleus works as a functional unit, serving as a relay station between the outer layers of the cerebellum and other parts of the brain.

# Cerebellar organization

In the cerebellum, some neuronal precursors can also migrate away from the germinal epithelium. These precursor cells, called **neuroblasts,** migrate to the outer surface of the developing cerebellum and form a new germinal zone, the **external granule layer,** near the outer boundary of the neural tube.

# Cerebellar organization

The inner compartment of the external granule layer contains postmitotic neuroblasts that are the precursors of the major neurons of the cerebellar cortex, the **granule neurons**.

# Cerebellar organization II

# Cerebellar organization

These granule neurons migrate back into the developing cerebellar white matter to produce a region called the internal granule layer.

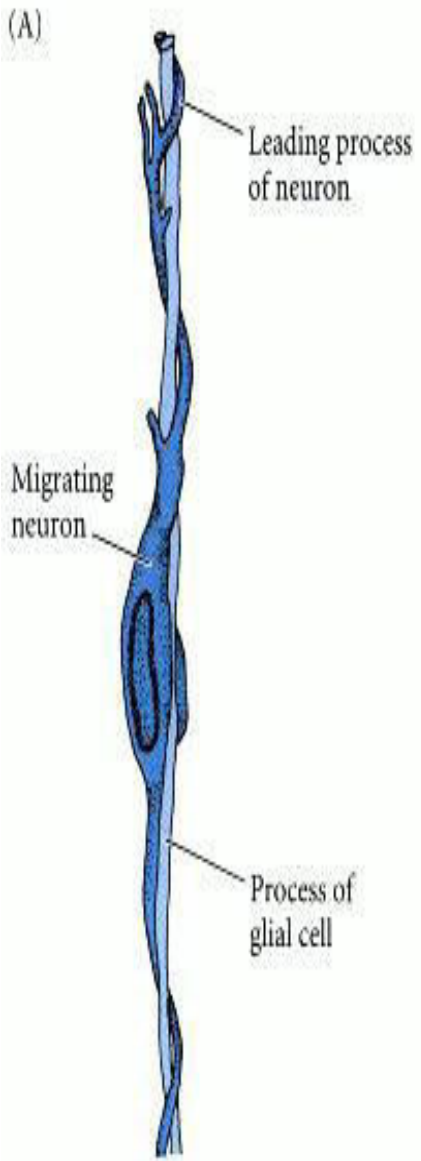
# Cerebellar organization

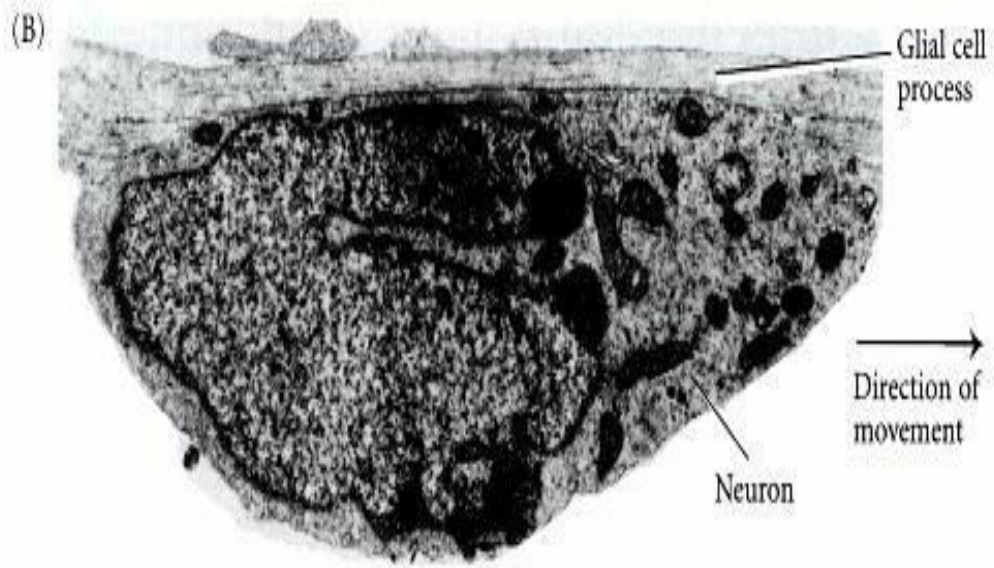
Meanwhile, the original ependymal layer of the cerebellum generates a wide variety of neurons and glial cells, including the distinctive and large **Purkinje neurons**.

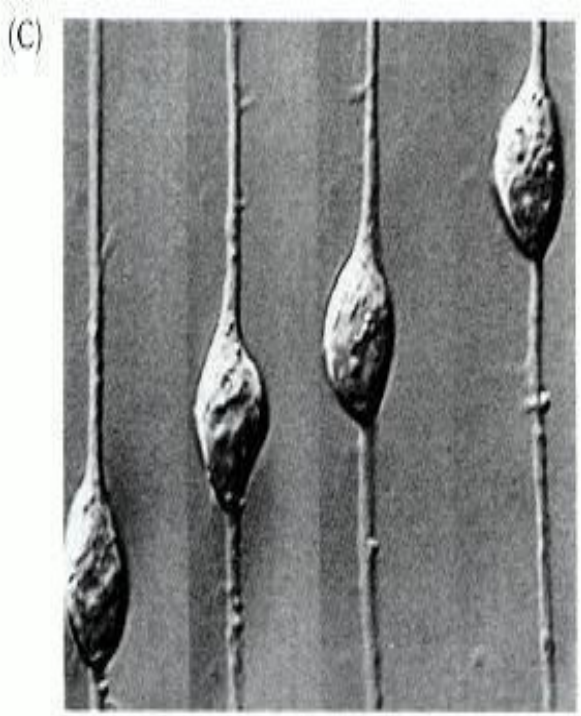
They secretes Sonic hedgehog, which sustains the division of granule neuron precursors in the external granule layer

# Cerebellar organization

One mechanism thought to be important for positioning young neurons within the developing mammalian brain is **glial guidance**.



# Neuronal Types

# Neuronal Types

The human brain consists of over **$10^{11}$** neurons associated with over **$10^{12}$** glial cells.

Those cells that remain integral components of the neural tube lining become ependymal cells. These cells can give rise to the precursors of neurons and glial cells.

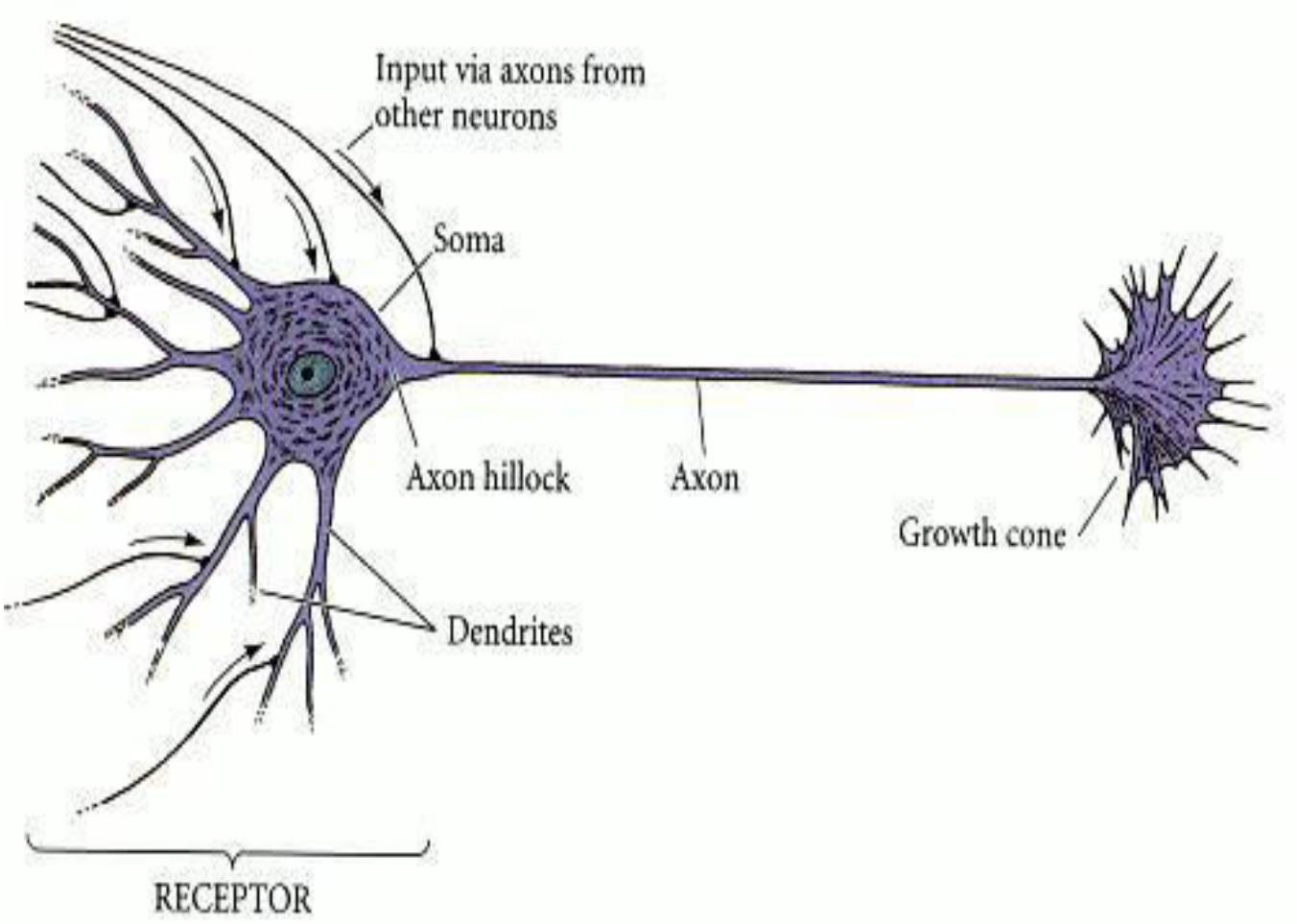
# Neuronal Types

The fine extensions of the neuron that are used to pick up electrical impulses from other cells are called **dendrites.** Some neurons develop only a few dendrites, whereas other cells (such as the Purkinje neurons) develop extensive dendritic trees.

# Development of Neurons

# Neuronal Types

Another important feature of a developing neuron is its axon (sometimes called a **neurite**).



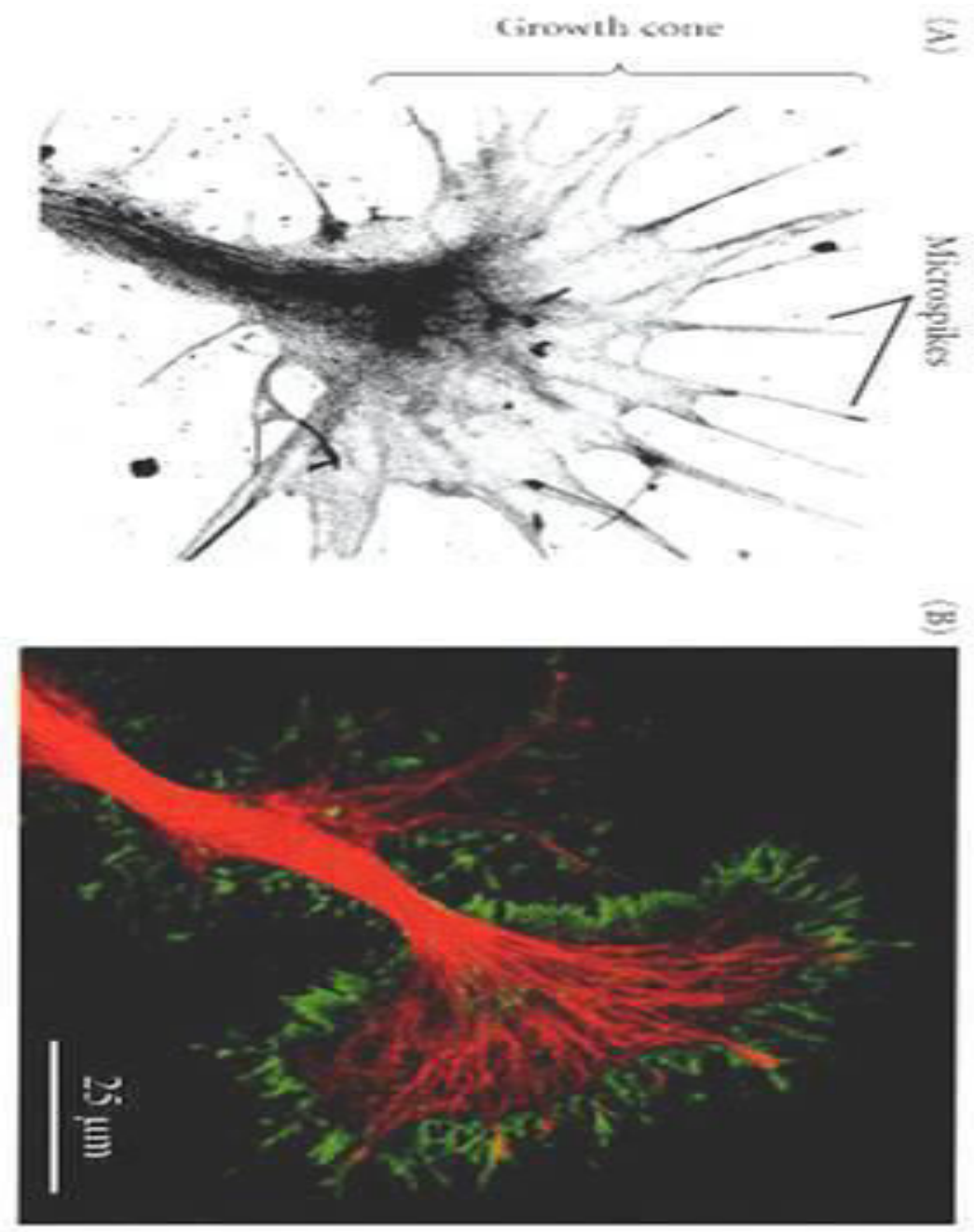
# Neuronal Types

Dendrites are often numerous and do not extend far from the neuronal cell body, or **soma**, axons may extend for several feet.

The Nerve outgrowth is led by the tip of the axon, called the growth cone.

# Neuronal Types

The growth cone moves by the elongation and contraction of pointed filopodia called Microspikes. These microspikes contain microfilaments, which are oriented parallel to the long axis of the axon.



# The Neural Crest

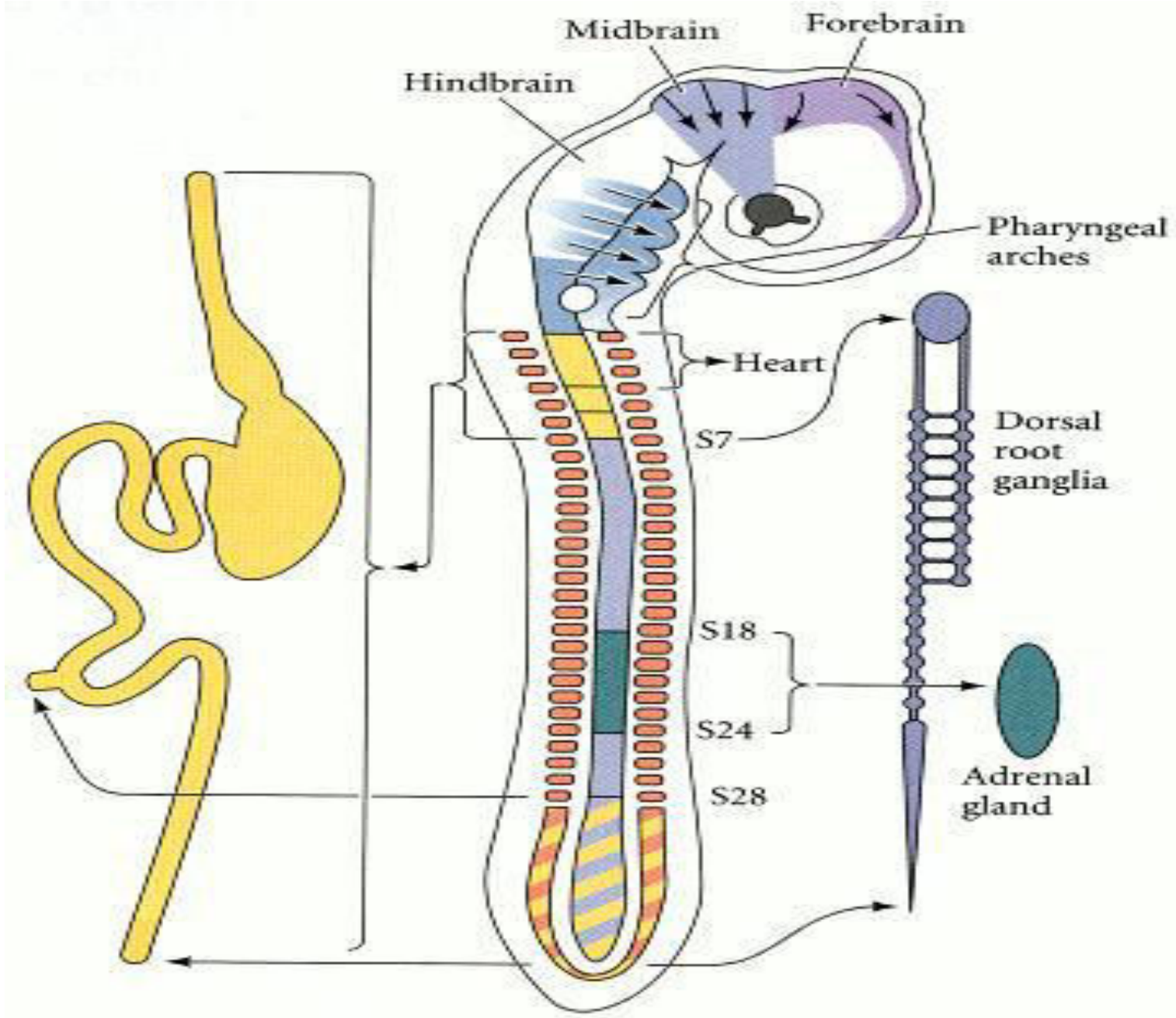
# The Neural Crest

A group of ectodermal cells that gives rise to the spinal ganglia and various structures of the autonomic nervous system.

The neural crest cells originate at the dorsal most region of the neural tube.

# The Neural Crest

The neural crest can be divided into four main functional (but overlapping) domains.



# The Neural Crest

The cranial (cephalic) neural crest, whose cells migrate dorso-laterally to produce the craniofacial mesenchyme that differentiates into the cartilage, bone, cranial neurons, glia, and connective tissues of the face.

# Major pathways in differentiation of neural crest

# The Neural Crest

The **trunk neural crest**, whose cells take one of two major pathways.

Neural crest cells that become the pigment-synthesizing melanocytes migrate dorso-laterally into the ectoderm and continue on their way toward the ventral midline of the belly.

# The Neural Crest

The second migratory pathway takes the trunk neural crest cells ventrolaterally through the anterior half of each sclerotome.

Sclerotomes are blocks of mesodermal cells, derived from somites, that will differentiate into the vertebral cartilage of the spine.

# The Neural Crest

The vagal and sacral neural crest, whose cells generate the parasympathetic (enteric) ganglia of the gut.

The vagal (neck) neural crest lies opposite chick somites 1-7, while the sacral neural crest lies posterior to somite 28.

# The Neural Crest

The cardiac neural crest is located between the cranial and trunk neural crests.

The cardiac neural crest cells can develop into melanocytes, neurons, cartilage, and connective tissue (of the third, fourth, and sixth pharyngeal arches).

# Summary
# Central Nervous System and Epidermis

# Summary
# Central Nervous System and Epidermis

- The neural tube forms from the shaping and folding of the neural plate. In primary neurulation, the surface ectoderm folds into a tube that separates from the surface. In secondary neurulation, the ectoderm forms a cord and then forms a cavity within it.

# Summary
# Central Nervous System and Epidermis

Primary neurulation is regulated by both intrinsic and extrinsic forces. Intrinsic wedging occurs within cells of the hinge regions to bend the neural plate. Extrinsic forces include the migration of the surface ectoderm towards the center of the embryo.

# Summary
# Central Nervous System and Epidermis

- Neural tube closure is also a mixture of extrinsic and intrinsic forces. In humans, if the neural tube fails to close various diseases can result.
- The neural crest cells arise at the lateral borders of the neural tube and surface ectoderm.

# Summary
## Central Nervous System and Epidermis

- There is a gradient of maturity in many embryos, especially those of amniotes. The anterior develops earlier than the posterior.
- The dorsal-ventral patterning of the neural tube is accomplished by proteins of the TGF-β family secreted from the surface ectoderm and roof of the neural tube.

# Summary Central Nervous System and Epidermis II

# Summary
# Central Nervous System and Epidermis

- The brain forms three primary vesicles: prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain).
- The brain expands through fluid secretion putting positive pressure on the vesicles.

# Summary
# Central Nervous System and Epidermis

- The neurons of the brain are organized into cortices (layers) and nuclei (clusters).
- New neurons are formed by mitosis in the neural tube. The germinal zone at the lumen of the neural tube is called the ventricular zone.

# Summary
# Central Nervous System and Epidermis

The cerebral cortex in humans has six layers, and the mantle zone is called the **Neocortex**. Cell fates are often fixed as they undergo their last division. Neurons derived from the same stem cell may end up in different functional regions of the brain.

# Summary
# Central Nervous System and Epidermis III

# Summary
## Central Nervous System and Epidermis III

- Dendrites receive signals from other neurons, while axons transmit them. The place where the signaling takes place (through the release of neurotransmitters) is called a **synapse**.
- Axons grow from the nerve cell body, or soma. They are led by the growth cone.

# Summary
# Central Nervous System and Epidermis III

The retina forms from the optic vesicle that extends from the brain.

Pax6 plays a major role in eye formation, and the down regulation of Pax6 by Sonic hedgehog in the center of the brain splits the eye-forming region of the brain in half.

# Summary
# Central Nervous System and Epidermis III

- The photoreceptor cells gather the light and transmit the impulse through interneurons to the retinal ganglion cells. The axons of the retinal ganglion cells form the optic nerve.

# Summary Central Nervous System and Epidermis IV

# Summary
# Central Nervous System and Epidermis
# IV

- Paracrine factors such as TGF- and FG**F7** are important in normal skin development.
- Cutaneous appendages hair, feathers, and scales are formed by epithelial-mesenchymal interactions between the epidermis and the dermal mesoderm.

# Summary
# Central Nervous System and Epidermis
# IIV

- The basal layer of the surface ectoderm becomes the stratum germinativum, or germinal layer of the skin. These cells divide to produce a stem cell and a cell committed to become an epidermal cell (keratinocyte). Stem cells appear to be able to make hair.

# Myogenesis:
# The Development of Muscle

# Myogenesis:
# The Development of Muscle

Paracrine factors instruct the **Myotome cells** to become muscles by inducing them to synthesize the **MyoD protein.**

Factors from the surrounding environment induce the Pax3 transcription factor.

# Myogenesis:
# The Development of Muscle

In the absence of other inhibitory transcription factors (such as those found in the sclerotome cells), Pax3 then activates the genes encoding two muscle-specific transcription factors, **Myf5** and **MyoD**.

# Myogenesis:
# The Development of Muscle

MyoD and Myf5 belong to a family of transcription factors called the myogenic bHLH (basic helix-loop-helix) proteins (sometimes also referred to as the MyoD family).

# Myogenesis II:
# The Development of Muscle

# Myogenesis:
# The Development of Muscle

The proteins of this family all bind to similar sites on the DNA and activate muscle-specific genes.

MyoD also directly activates its own gene.

# Myogenesis:
# The Development of Muscle

While Pax3 is found in several other cell types, the myogenic bHLH proteins are specific for muscle cells. Any cell making a myogenic bHLH transcription factor such as MyoD or Myf5 is committed to becoming a muscle cell.

# Myogenesis:
# The Development of Muscle

Transfection of genes encoding any of these myogenic proteins into a wide range of cultured cells converts those cells into muscles.

# Muscle cell fusion

# Muscle cell fusion

The myotome cells producing the myogenic bHLH proteins are the myoblasts- committed muscle cell precursors.

These cells align together and fuse to form the multinucleated myotubes**, a** characteristic of muscle tissue.

# Muscle cell fusion

Muscle cell fusion begins when the myoblasts leave the cell cycle. As long as particular growth factors (particularly fibroblast growth factors) are present, the myoblasts will proliferate without differentiating.

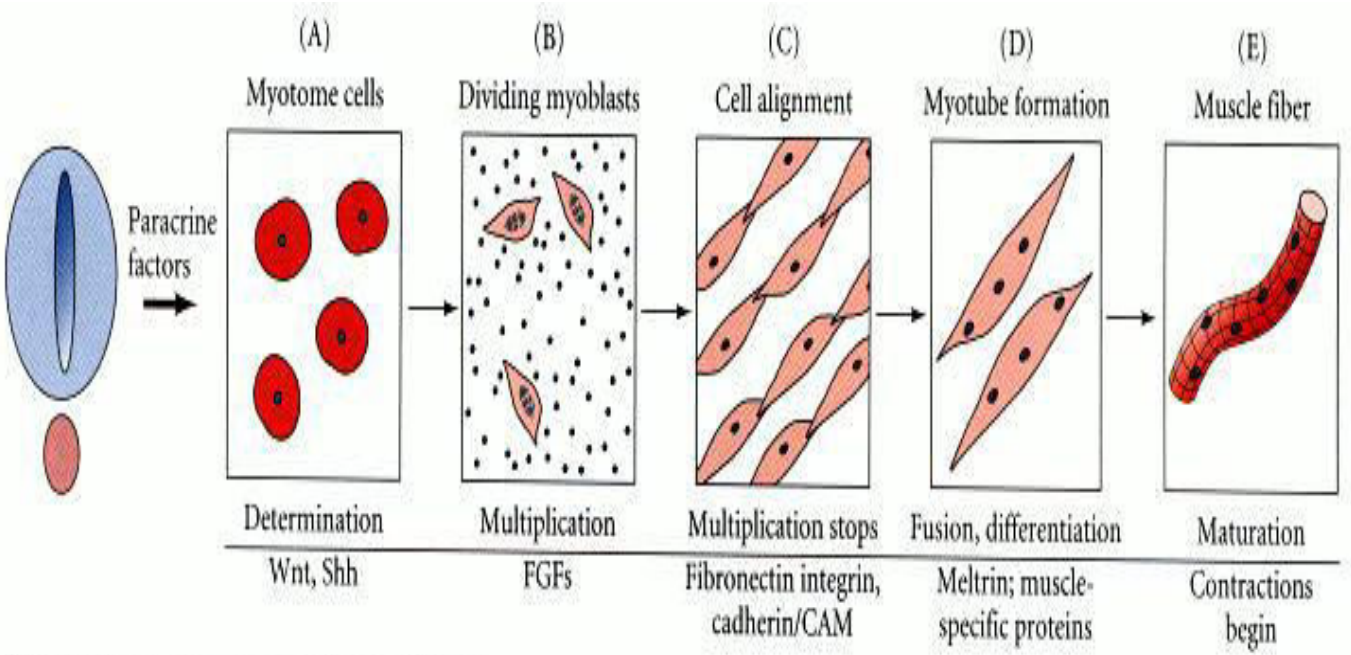
# Steps in Muscle cell fusion

# Muscle cell fusion

When these factors are depleted, the myoblasts stop dividing, secrete **fibronectin** onto their extracellular matrix, and bind to it through **5β1 integrin**, their major fibronectin receptor.

# Muscle cell fusion

The second step is the alignment of the myoblasts together into chains.






# Muscle cell fusion

The third step is the cell fusion event itself. As in most membrane fusions, calcium ions are critical, and fusion can be activated by **calcium ionophores, such as A23187**, that carry calcium ions across cell membranes.

# Muscle cell fusion

Fusion appears to be mediated by a set of metalloproteinases called **meltrins**. These proteins were discovered during a search for myoblast proteins that would be homologous to **fertilin**, a protein implicated in sperm-egg membrane fusion.

# Osteogenesis:
# The Development of Bones

# Osteogenesis:
# The Development of Bones

Some of the most obvious structures derived from the paraxial mesoderm are bones.

There are three distinct lineages that generate the skeleton.

# Osteogenesis:
# The Development of Bones

- The somites generate the axial skeleton.
- The lateral plate mesoderm generates the limb skeleton.
- The cranial neural crest gives rise to the branchial arch and craniofacial bones and cartilage.

# Osteogenesis II:
# The Development of Bones

# Osteogenesis: The Development of Bones

There are two major modes of bone formation, or **osteogenesis**, and both involve the transformation of a preexisting mesenchymal tissue into bone tissue.

# Osteogenesis: The Development of Bones

- The direct conversion of mesenchymal tissue into bone is called **intramembranous ossification**. This process occurs primarily in the bones of the skull.

# Osteogenesis: The Development of Bones

- In other cases, the mesenchymal cells differentiate into cartilage, and this cartilage is later replaced by bone. The process by which a cartilage intermediate is formed and replaced by bone cells is called endochondral ossification.

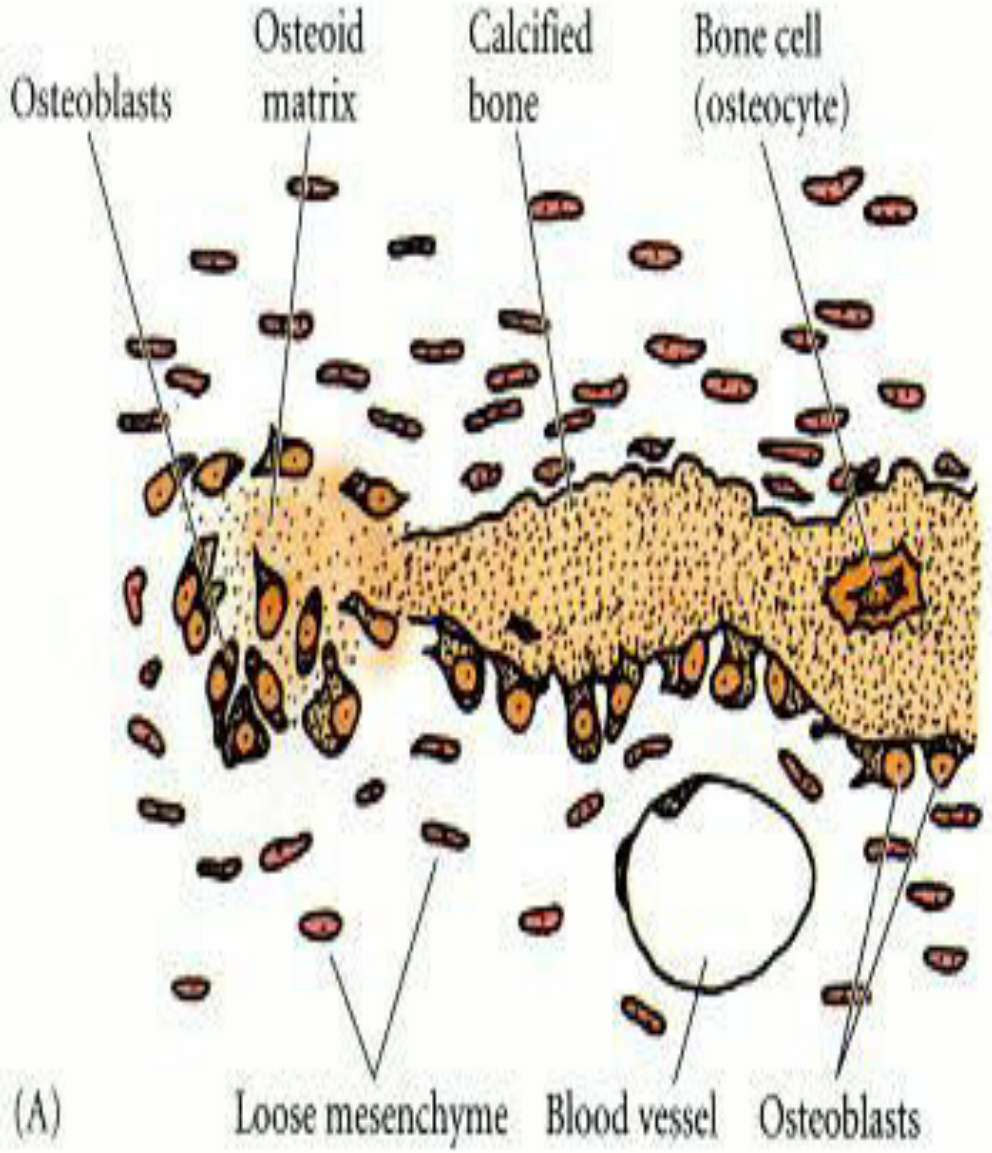
# Intramembranous ossification

# Intramembranous ossification

Intramembranous ossification is the characteristic way in which the flat bones of the skull and the turtle shell are formed. During intramembranous ossification in the skull, neural crest derived mesenchymal cells proliferate and condense into compact nodules.

# Intramembranous ossification

Some of these cells develop into capillaries; others change their shape to become osteoblasts, committed bone precursor cells.






# Intramembranous ossification

The osteoblasts secrete a **collagen-proteoglycan matrix** that is able to bind calcium salts. Through this binding, the prebone (osteoid) matrix becomes calcified.

In most cases, osteoblasts are separated from the region of calcification by a layer of the osteoid matrix they secrete.

# Intramembranous ossification II

# Intramembranous ossification

Occasionally, though, osteoblasts become trapped in the calcified matrix and become osteocytes bone cells. As calcification proceeds, bony spicules radiate out from the region where ossification began.

# Intramembranous ossification

Furthermore, the entire region of calcified spicules becomes surrounded by compact mesenchymal cells that form the Periosteum (a membrane that surrounds the bone).

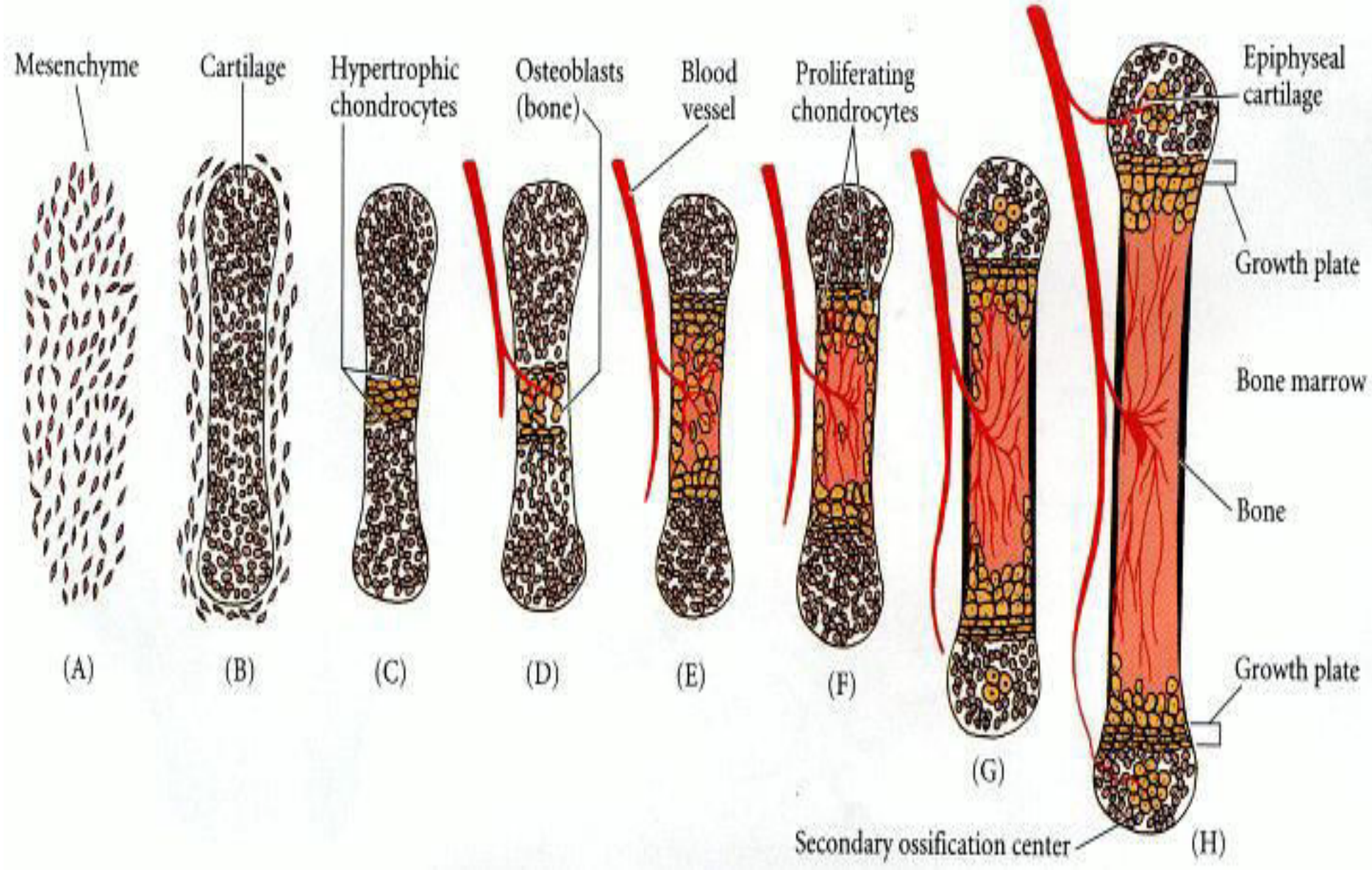
# Endochondral ossification

# Endochondral ossification

Endochondral ossification involves the formation of cartilage tissue from aggregated mesenchymal cells, and the subsequent replacement of cartilage tissue by bone.

# Endochondral ossification

The process of Endochondral ossification can be divided into five stages.



# Endochondral ossification

First, the mesenchymal cells are committed to become cartilage cells. This commitment is caused by paracrine factors that induce the nearby mesodermal cells to express two transcription factors, Pax1 and **Scleraxis**.

# Endochondral ossification II

# Endochondral ossification

Pax1 and Scleraxis are the transcription factors are

thought to activate cartilage-specific genes.

# Endochondral ossification

Thus, Scleraxis is expressed in the mesenchyme from the sclerotome, in the facial mesenchyme that forms cartilaginous precursors to bone, and in the limb mesenchyme

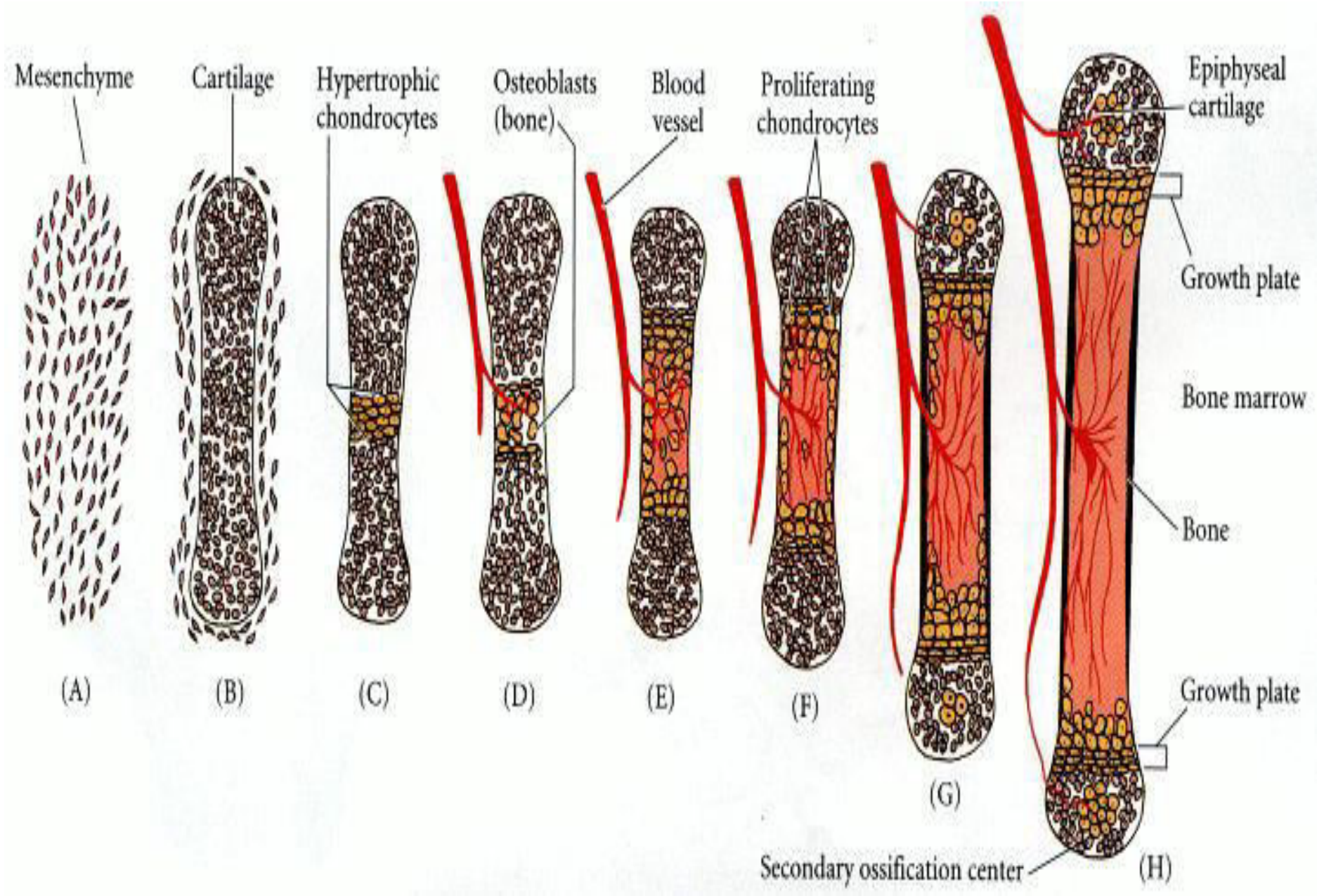
# Endochondral ossification

During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells. N-cadherin appears to be important in the initiation of these condensations.

# Endochondral ossification IiI

# Endochondral ossification III

During the second phase of endochondral ossification, the committed mesenchyme cells condense into compact nodules and differentiate into **chondrocytes**, the cartilage cells.



# Endochondral ossification III

In humans, the *SOX9* gene, which encodes a DNA-binding protein, is expressed in the precartilaginous condensations. Mutations of the *SOX9* gene cause camptomelic dysplasia, a rare disorder of skeletal development that results in deformities of most of the bones of the body.

# Endochondral ossification III

During the third phase of endochondral ossification, the chondrocytes proliferate rapidly to form the model for the bone. As they divide, the chondrocytes secrete a cartilage-specific extracellular matrix.

# Endochondral ossification IV

# Endochondral ossification IV

In the fourth phase, the chondrocytes stop dividing and increase their volume dramatically, becoming hypertrophic chondrocytes. These large chondrocytes alter the matrix they produce (by adding collagen X and more fibronectin) to enable it to become mineralized by calcium carbonate.

# Endochondral ossification IV

The fifth phase involves the invasion of the cartilage model by blood vessels.

**The hypertrophic chondrocytes die by apoptosis.**

This space will become **bone marrow**. As the cartilage cells die, a group of cells that have surrounded the cartilage model differentiate into osteoblasts.

# Endochondral ossification IV

The osteoblasts begin forming bone matrix on the partially degraded cartilage. Eventually, all the cartilage is replaced by bone. Thus, the cartilage tissue serves as a model for the bone that follows.

# Osteoclasts

# Osteoclasts

Osteoclasts are probably derived from the same precursors as macrophage blood cells, and they dissolve both the inorganic and the protein portions of the bone matrix.

# Osteoclasts

Each osteoclast extends numerous cellular processes into the matrix and pumps out hydrogen ions onto the surrounding material, thereby acidifying and solubilizing it.

# Osteoclasts

The blood vessels also import the blood-forming cells that will reside in the marrow for the duration of the organism's life. The number and activity of osteoclasts must be tightly regulated.

# Osteoclasts II

# Osteoclasts

If there are too many active osteoclasts, too much bone will be dissolved, and osteoporosis will result. Conversely, if not enough osteoblasts are produced, the bones are not hollowed out for the marrow, and osteopetrosis results.

# Osteoclasts

## Osteopetrosis or Marble bone disease

Osteopetrosis is a bone disease that makes bones abnormally dense and prone to breakage (fracture).

This disorder is characterized by increased bone density and abnormal bone growth.